

WHO information for molecular diagnosis of influenza virus - update

It is strongly recommended that all un-subtypable influenza A specimens should be immediately sent for diagnosis and further characterization to one of the six WHO Collaborating Centres for Reference & Research on Influenza¹.

March 2014

Revision May 2015:

- ***Annex 2.D, protocol 1 updated (pages 30-35)***
- ***Annex 2.E, protocol 4 added (pages 50-53)***

This document provides information on molecular diagnostic protocols updated as of the above date for surveillance of influenza viruses in humans.

This is a revision of the document published on the WHO website in November 2012 with updated molecular protocols. The protocols are in the annexes below:

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¹ http://www.who.int/influenza/gisrs_laboratory/collaborating_centres/list/en/index.html

Specimens

The most appropriate specimens for the diagnosis of influenza are upper respiratory tract specimens. Samples should be taken from the deep nostrils (nasal swab), throat (oropharyngeal swab) and nasopharynx (nasopharyngeal swab). Nasopharyngeal aspirate and bronchial aspirate are also useful. Appropriate precautions should be taken in collecting specimens since this may expose the collector to respiratory secretions from patients.

Laboratory tests

Molecular diagnostic techniques are rapid and sensitive methods for the detection and identification of influenza viruses, both for clinical samples and isolates. The reverse-transcription polymerase chain reaction (RT-PCR) allows template viral RNA to be reverse transcribed producing complementary DNA (cDNA) which can then be amplified and detected.

Protocols for influenza RT-PCR detection and subtyping of influenza are outlined below.

In addition to RT-PCR, other laboratory techniques are available for the detection, identification and characterization of influenza virus including **virus isolation in cell culture or fertilized chicken eggs**, characterization of the isolated virus by haemagglutination inhibition (HAI) testing, immunofluorescence detection of the virus in clinical specimens or isolates, rapid antigen tests and other molecular techniques.

Detailed and updated information on these methodologies are included in the “Manual for the laboratory diagnosis and virological surveillance of influenza” recently published by WHO.²

² http://www.who.int/influenza/gisrs_laboratory/manual_diagnosis_surveillance_influenza/en/index.html

Molecular diagnostics

The RT-PCR technique is used as a rapid and sensitive method for the detection of influenza viruses in both clinical samples and isolates.

The use of different target gene assays in the RT-PCR is most appropriate for correct identification of this virus. The following gene targets, among others, are important:

- type A influenza matrix gene;
- haemagglutinin gene specific for influenza A subtypes:
 - A(H1N1)2009 virus (formerly pandemic A(H1N1) 2009), A(H3N2), former seasonal A(H1N1) and the highly pathogenic avian influenza A(H5N1) virus;
- haemagglutinin gene specific for influenza B virus.

The following protocols are currently available:

- influenza A type-specific conventional and realtime-PCR (see Annexes 1 and 2);
- A(H1N1)2009 virus specific conventional and realtime-PCR (see Annexes 1 and 2);
- CDC realtime RT-PCR (rRT-PCR) protocol for the detection and characterization of A(H1N1) 2009;³
- former seasonal influenza A(H1N1), influenza A(H3N2), influenza B and avian influenza A(H5N1) realtime RT-PCR (see Annexes 1 and 2);
- influenza B lineages conventional and realtime RT-PCR (see Annexes 1 and 2);
- A(H7N9) conventional and realtime PCR (see Annex 2).

For current PCR protocols for H7 viruses please contact the Department of Virology, Erasmus MC Rotterdam, Netherlands at <http://www.virology.nl/>

The majority of the molecular diagnostic procedures described in this document were developed by members of the WHO Expert Committee on influenza PCR. Members of this Committee are from the WHO H5 Reference laboratories.⁴

Interpretation of PCR results

- PCR — A sample is considered positive if results from tests using two different PCR targets (e.g. primers specific for universal M gene and A(H1N1)2009 haemagglutinin gene) are positive. If RT-PCR for multiple haemagglutinin (HA) targets (e.g. A(H3), and A(H1N1)2009) give positive results in the same specimen, the possibility of PCR contamination should first be excluded by repeating the PCR procedure using a new RNA extract from the original specimen or an RNA extract from another specimen. If repeated positive results for multiple HA targets are obtained, this raises the possibility of coinfection, which should be confirmed by sequencing or virus culture.
- CDC realtime-PCR assays — Results should be interpreted as described in the CDC H1N1 realtime assay manual.²
- A negative PCR result does not rule out that a person may be infected with an influenza virus. Results should be interpreted in conjunction with the available clinical and epidemiological information. Specimens from patients whose PCR results are negative

³ <http://www.who.int/csr/resources/publications/swineflu/realtimeptpcr/en/index.html>

⁴ http://www.who.int/influenza/gisrs_laboratory/h5_reflabs/list/en/index.html

but for whom there is a high suspicion of influenza infection should be further investigated and tested by other methods such as virus culture or serology.¹

Referral for further characterization

Specimens with laboratory results indicative of influenza A that are un-subtypeable (i.e. negative for influenza A(H1), A(H3) and A(H5)) should be forwarded to a WHO CC as soon as possible for confirmation.

Laboratories with no virus isolation capacity or required biosafety containment levels should forward the specimens to a WHOCC.

Standard and relevant IATA regulations for influenza specimen storage, packaging and shipping practices should be followed.⁵

To allow optimal recovery of virus isolates from clinical specimens known to be influenza positive based on real-time RT-PCR, those with Ct values of up to 30 should be shipped to a WHO CC. Exceptions to this guidance are when specimens are associated with severe or fatal disease but have Ct values in the range 31–40, and specimens that are unsubtypeable (i.e. are influenza A positive but not HA positive for a seasonal influenza subtype). All unsubtypeable specimens should be shipped to a WHO CC as soon as possible. When urgent detection/diagnosis for clinical needs is required, WHO CCs can provide this service, but possibly not in the most timely manner due to shipment times. Ideally, such a service may be provided by accredited hospital/private laboratories or the National Influenza Centre (NIC) of the country itself or a close neighbour.

Biosafety

Diagnostic laboratory work on clinical specimens and virus isolation from patients who are suspected of being infected with seasonal influenza virus should be conducted in BSL2 containment conditions with the use of appropriate personal protective equipment (PPE). All clinical specimen manipulations should be done inside a certified biosafety cabinet (BSC). Please refer to the WHO *Laboratory biosafety manual*, 3rd edition.⁶

Testing algorithms

The overall approach to influenza virus detection by RT-PCR should be considered in the context of the national situation; e.g. How many specimens can be handled (throughput), what gene sequence to target for RT-PCR, and whether to use concurrent or sequential testing for RT-PCR of M and HA genes.

Good laboratory practices

Standard protocols for all procedures should be in place and reviewed regularly. Ensuring that the recommended reagents are used and handled properly is critical, as reactions are complex and problems with a single reagent can have a significant effect on the results.

⁵ http://www.who.int/csr/resources/publications/swineflu/storage_transport/en/index.html

⁶ http://www.who.int/csr/resources/publications/biosafety/WHO_CDS_CSR_LYO_2004_11/en/

Validation

All protocols should always be validated in each laboratory to ensure adequate specificity and sensitivity.

Quality assurance

Standard quality assurance protocols including the use of appropriate controls and good laboratory practices should be in place. Participation in the NIC evaluation exercises (external quality assessment programme) is highly recommended to confirm that laboratories are achieving an adequate level of sensitivity and specificity in their tests.

Training of personnel

Familiarity with protocols and experience in correct interpretation of results are cornerstones for successful execution of the diagnostic tests.

Facilities and handling areas

Specimen and reagent handling facilities (including cold chains) with appropriate separation for different steps of RT-PCR must be in place to prevent cross-contamination. Facilities and equipment should meet the appropriate biosafety level. RT-PCR should be performed in a space separated from that used for virus isolation techniques.

Equipment

Equipment should be used and maintained according to the manufacturer's recommendations.

Annex 1: Conventional RT-PCR protocols

A. Conventional RT-PCR analyses for the matrix gene of influenza type A viruses⁷

The following protocols are for conventional RT-PCR and gel electrophoresis of PCR products to detect influenza type A viruses (all subtypes) in specimens from humans. These protocols have been shown to be widely effective for the identification of influenza type A viruses when used with the reagents and primers indicated. For laboratories that have concerns about identifying currently circulating viruses, it is recommended that they contact one of the WHO H5 Reference laboratories⁸ for diagnosis of influenza infection or one of the WHO CCs⁹ for assistance in identifying the optimal primers to be used.

Materials required

- QIAamp[®] Viral RNA Mini Kit (QIAGEN[®], Cat. No. 52904. Other extraction kits can be used after proper evaluation)
- OneStep RT-PCR Kit (QIAGEN[®], Cat. No. 210212)
- RNase Inhibitor 20U/μl (Applied Biosystems, Cat. No. N8080119)
- RNase-free water
- Ethanol (96–100%)
- Microcentrifuge (adjustable up to 13 000 rpm)
- Adjustable pipettes (10, 20, 200, and 100 μl)
- Sterile, RNase-free pipette tips with aerosol barrier
- Vortex
- Microcentrifuge tubes (0.2, 1.5 ml)
- Thermocycler (PCR machine)
- Primer sets
- Positive control (may be obtained upon request from a WHOCC)

Primer sequences

Type/subtype	Gene fragment	Primer	Sequence
Influenza type A	Matrix (M)	M30F2/08 M264R3/08	ATGAGYCTTYTAACCGAGGTCGAAACG TGGACAAANCGTCTACGCTGCAG

Expected PCR product size is 244 bp

Procedure

1. Extract viral RNA from clinical specimen with QIAamp Viral RNA Mini Kit or equivalent extraction kit, according to manufacturer's instructions.
2. Perform one-step RT-PCR:

⁷ WHO Collaborating Centre for Reference and Research on Influenza. National Institute of Infectious Diseases. Gakuen 4-7-1, Musashi-Murayama-shi, Tokyo 208-001, Japan. Email: todagiri@nih.go.jp
<http://idsc.nih.go.jp/>

⁸ http://www.who.int/influenza/gisrs_laboratory/h5_reflabs/en/

⁹ http://www.who.int/influenza/gisrs_laboratory/collaborating_centres/list/en/

- Take out the reagents from storage and thaw them at room temperature. After they are thawed out, keep them on ice.
- Preparation of master mix (**operate on ice**)
 - Add the following to microcentrifuge tubes and mix gently by pipetting the master mix up and down ten times. (Note: To avoid localized differences in salt concentration, it is important to mix the solutions completely before use).

Reaction without Q-Solution

Reagent	Volume (μl)
Water (molecular grade)	9.5
5X QIAGEN® RT-PCR buffer	5.0
dNTP mix (containing 10mM of each dNTP)	1.0
Forward primer (10μmol/l)	1.5
Reverse primer (10μmol/l)	1.5
QIAGEN® OneStep RT-PCR Enzyme mix (5U/μl)	1.0
RNase Inhibitor (20U/μl)	0.5
Total volume	20.0

- Dispense 20μl of the master mix to each PCR reaction tube.
- Add 5μl sample RNA to the master mix. For control reactions, use 5μl of distilled water for negative control and 5μl of appropriate viral RNAs for positive control.
- Program the thermal cycler according to thermal cycling conditions.
- Start the RT-PCR program while PCR tubes are still on ice. Wait until the thermal cycler has reached 50° C, then place the PCR tubes in the thermal cycler.

Thermal cycling conditions

Type of cycle	Temperature (° C)	Time (minute:second)	No. of cycles
Reverse transcription	50	30:00	1
Initial PCR activation	95	15:00	1
Three step cycling:			
Denaturation	94	0:30	45
Annealing	50	0:30	
Extension	72	1:00	
Final Extension	72	10:00	1

3. Agarose gel electrophoresis of RT-PCR products.

Prepare agarose gel, load PCR products and molecular weight marker, and run according to standard protocols. Visualize presence of marker under UV light. An example of the material required and the procedure is given below.

Materials required

- Agarose gel casting tray and electrophoresis chamber
- Power supply and electrode leads
- UV light box ($\lambda = 302\text{nm}$)
- Camera and Polaroid® film or use any digital gel documentation system
- Adjustable pipettes
- 2% agarose gel in 1X TAE buffer
- 1X TAE buffer
- Ethidium bromide (10mg/ml)
- 6x Gel loading buffer (GLB)
- Molecular weight marker

Procedure

A) Casting the agarose gel:

- i) Place a gel-casting tray onto a gel-casting base. Insert a comb and level the base.
- ii) Prepare 2% agarose by weighing out 4g of agarose powder and dissolving it in 200ml 1X TAE buffer. Dissolve the agar by heating in microwave oven.
- iii) Cool the melted agarose to about 60° C, then add 10 μ l of ethidium bromide.
- iv) Pour the melted agarose into the gel-casting tray.
- v) Allow the gel to solidify at room temperature.
- vi) Remove the comb from the frame.
- vii) Place the tray into the electrophoresis chamber with the wells at the cathode side.
- viii) Fill the buffer chamber with 1X TAE at a level that can cover the top of the gel.

B) Sample loading:

- i) Add 5 μ l of the gel loading buffer to each PCR tube.
- ii) Load molecular weight marker to the first well of the agarose gel.
- iii) Pipette 15 μ l of the PCR product in gel loading buffer to the gel.
- iv) Close the lid on the chamber and attach the electrodes. Run the gel at 100V for 30–35 minutes.
- v) Visualize the presence of marker and PCR product bands with a UV light.
- vi) Document the gel picture by photographing it.

Interpretation of results

The size of PCR products obtained should be compared with the expected product size. Tests should always be run with a positive control.

B. Conventional one step RT-PCR for A(H1N1)2009 HA gene¹⁰

The protocols and primers for conventional RT-PCR to detect A(H1N1)2009 viruses in specimens from humans are given below. It is recommended that laboratories having concerns about identifying currently circulating viruses should contact one of the WHO H5 Reference laboratories or one of the WHO CCs for assistance in identifying the optimal primers to be used.

These assays were validated on the following working platforms:

GeneAmp PCR system 9700 (Applied Biosystems)

Veriti 96-well thermal cycler (Applied Biosystems)

Materials required

- QIAamp Viral RNA Mini Kit (QIAGEN®, Cat. No. 52904)
- QIAGEN OneStep RT-PCR kit (QIAGEN®, Cat. No. 210212)
- RNase inhibitor 20U/μl (Applied Biosystems, Cat. No. N808-0119)
- Ethanol (96–100%)
- Microcentrifuge (adjustable, up to 13 000rpm)
- Adjustable pipettes (10, 20, 100, 200μl)
- Sterile, RNase-free pipette tips with aerosol barrier
- Vortex
- Microcentrifuge tubes (0.2, 1.5ml)
- Thermocycler (GeneAmp PCR system 9700, Applied Biosystems or Veriti 96-well thermal cycler, Applied Biosystems)
- Positive control (Swine influenza A virus A/SW/HK/PHK1578/03 or A/California/04/2009) (Available upon request from Hong Kong University)
- Primer set

Primers

Type/subtype	Gene fragment	Primer	Sequence
Influenza A(H1N1)2009 virus	HA	HKU-SWF HKU-SWR	GAGCTCAGTGTCATCATTTGAA TGCTGAGCTTTGGGTATGAA

Expected PCR product size is 173 bp.

Procedure

1. Extract viral RNA from clinical specimen with QIAamp Viral RNA Mini Kit or equivalent extraction kit according to manufacturer's instructions.

¹⁰ Department of Microbiology, Faculty of Medicine, University of Hong Kong, University Pathology Building Queen Mary Hospital, Hong Kong Special Administrative Region of China.

2. Prepare master mixture for RT-PCR as below:

Reagent	Volume (μ l)	Final concentration
Water (molecular grade)	7.4	
5X PCR buffer (kit)	4.0	1X
dNTPs (kit)	0.8	400 μ M of each dNTP
5 μ M primer : HKU-SWF	2.4	0.6 μ M
5 μ M primer : HKU-SWR	2.4	0.6 μ M
Rnase Inhibitor (20U/ μ l)	0.2	4 U
Enzyme mix (kit)	0.8	-
Total	18.0	

3. Dispense 18 μ l of master mix into each test tube.

4. Add 2 μ l of purified RNA to the above reaction mix.

5. Set the following RT-PCR conditions:

Step	Temperature ($^{\circ}$ C)	Time (minute:second)	No. of cycles
Reverse transcription	50 95	30:00 15:00	1
Denaturation	94	0:30	40
Annealing	57	0:30	
Extension	72	0:20	
Post-PCR extension	72	7:00	1
Post-run	4	Hold	

6. Prepare 2% agarose gel, load PCR products and molecular weight markers, and run according to standard protocols. Visualize presence of marker and PCR product bands under UV light.

Interpretation of results

The expected size of PCR products for influenza H1 is 173 bp. This assay can specifically detect samples with A(H1N1)2009, but not those with former seasonal A(H1N1). RNA samples extracted from seven former seasonal A(H1N1), two A(H3N2), one human A(H5N1), seven avian influenza viruses (HA subtypes 4, 5, 7, 8, 9, and 10) and >150 nasopharyngeal aspirate samples from patients with other respiratory diseases were all negative in the assay. It should be noted that these assays can detect A(H1N1)2009 and some other swine H1 viral sequences. One of the positive controls recommended in this assay is a swine H1 virus isolated in Hong Kong. The absence of the correct PCR products (i.e. a negative result) does not rule out the presence of influenza virus. Results should be interpreted together with the available clinical and epidemiological information.

C. Conventional one step RT-PCR for A(H5N1) HA gene

Protocol 1:¹¹

This conventional RT-PCR protocol and primer set are designed to detect highly pathogenic H5N1 viruses in human specimens. Suitable biosafety precautions should be made for handling suspected H5 samples. Laboratories that have concerns about identifying currently circulating H5 viruses should contact one of the WHO Collaborating Centres or WHO H5 Reference laboratories for assistance in identifying the optimal primers to be used.

Materials required

- QIAamp Viral RNA Mini Kit (QIAGEN® Cat. No. 51104) or equivalent extraction kit
- QIAGEN OneStep RT-PCR kit (QIAGEN®, Cat. No. 210212)
- RNase inhibitor 20U/ml, (Applied Biosystems Cat. No. N808-0119)
- Ethanol (96–100%)
- Microcentrifuge (adjustable, up to 13 000 rpm)
- Adjustable pipettes (10, 20, 100, 200µl)
- Sterile, RNase-free pipette tips with aerosol barrier
- Vortex
- Microcentrifuge tubes (0.2, 1.5ml)
- Thermocycler (Applied Biosystems 9700)
- Positive control (Can be obtained from HKU)
- Primer set

Primers

Type/subtype	Gene fragment	Primer	Sequence
Influenza A(H5N1) virus	HA	H5-918F H5-1166R	CCARTRGGKGCKATAAAAYTC GTCTGCAGCRTAYCCACTYC

Expected PCR product size is 249 bp.

Procedure

1. Extract viral RNA from clinical specimen with QIAamp Viral RNA Mini Kit or equivalent extraction kit according to manufacturer's instructions.

¹¹ Department of Microbiology, Faculty of Medicine, University of Hong Kong, University Pathology Building Queen Mary Hospital, Hong Kong Special Administrative Region of China.

2. Prepare master mixture for one step RT-PCR as below:

Reagent	Volume(μ l)	Final concentration
Water (molecular grade)	1.7	
5X PCR buffer (kit)	2.0	1X
dNTPs (kit)	0.4	400 μ M of each dNTP
5X Q-sol (kit)	2.0	1X
5 μ M primer : H5- 918F	1.2	0.5 μ M
5 μ M primer : H5- 1166R	1.2	0.5 μ M
Enzyme mix (kit)	0.4	-
Rnase Inhibitor (20U/ μ l)	0.1	10 U
Total	9.0	

3. Dispense 9 μ l of master mix into each test tube.

4. Add 1 μ l of purified RNA to the above reaction mix.

5. Set the follow RT-PCR conditions:

Step	Temperature ($^{\circ}$ C)	Time (minute:second)	No. of cycles
Reverse transcription	50	30:00	1
	95	15:00	
Denaturation	94	0:30	45
Annealing	55	0:30	
Extension	72	0:30	
Post-PCR extension	72	7:00	1
Post-run	4	∞	

6. Prepare 2% agarose gel, load PCR products and molecular weight marker, and run according to standard protocols. Visualize presence of marker and PCR product bands under UV light.

Interpretation of results

The size of PCR products obtained should be compared with the expected product size. If the test is run without a positive control, products must be confirmed by sequencing and comparison with available sequences. The absence of the correct PCR products (i.e. a negative result) does not rule out the presence of influenza virus. Results should be interpreted together with the available clinical and epidemiological information.

Real-time RT-PCR format

The assay can be converted into a real-time RT-PCR format by including a hydrolysis probe (H5-Probe: FAM-ACCATKCCYTGCCAYCCYCCYTCT-3BHQ1).

Validation of PCR

The following highly pathogenic H5N1 viruses were found to be positive in the assay:

Clade	Virus
0	A/Hong Kong/483/1997
2.1.1	A/chicken/Hong Kong/YU324/2003
2.2	A/bar-headed goose/Qinghai/5/2005
2.3.1	A/duck/Hunan/139/2005
2.3.2	A/grey heron/Hong Kong/837/2004
2.3.2.1	A/grey heron/Hong Kong/779/2009
2.3.3	A/duck/Guiyang/3242/2005
2.3.4	A/chestnut-munia/Hong Kong/2442/2007
2.3.4.2	A/goose/Yunnan/6193/2006
2.3.4.3	A/blue-magpie/Hong Kong/1993/2007
2.4	A/chicken/Yunnan/6957/2003
2.5	A/goose/Shantou/239/2006
3	A/chicken/Hong Kong/YU562/01
4	A/chicken/Hong Kong/96.1/02
5	A/duck/Guangxi/1378/2004
7	A/goose/Yunnan/3315/2005
8	A/chicken/Hong Kong/86.3/2002
9	A/peregrine falcon/Hong Kong/D0028/2004

Protocol 2:¹²

The protocols and primers for conventional RT-PCR to detect highly pathogenic H5N1 viruses in specimens from humans are given below. It is recommended that laboratories having concerns about identifying currently circulating viruses should contact one of the members of the WHO Expert Committee on influenza PCR or one of the WHO CCs for assistance in identifying the optimal primers to be used.

Primers and probes

Type/subtype	Gene fragment	Primer	Sequence
Influenza A H5N1 virus	HA	H5-248-270F H5-671-647R	GTGACGAATTCATCAATGTRCCG CTCTGGTTTAGTGTTGATGTYCCAA

Expected PCR product size is 424 bp.

Procedure

Follow the same procedure and steps described on page 5 for detection of the universal M gene RT-PCR protocol developed by the US National Institute of Infectious Diseases (NIID).

Interpretation

The size of PCR products obtained should be compared with the expected product size. Tests should always be run with a positive control.

¹² WHO Collaborating Centre for Reference and Research on Influenza and WHO H5 Reference Laboratory at National Institute of Infectious Diseases (NIID). Gakuen 4-7-1, Musashi-Murayama-shi, Tokyo 208-001, Japan.

D. Influenza B lineage-specific one-step conventional RT-PCR protocol¹³

This conventional RT-PCR protocol and primer set are designed to detect influenza B Victoria/2/87 and B Yamagata/16/88 lineages.

Materials required

- QIAGEN® RNeasy Mini Kit (Cat. No. 74104 or 74106)
- QIAGEN® OneStep RT-PCR Kit (Cat. No. 210212)
- Forward and reverse primers
- Molecular grade sterile distilled water (RNase and DNase free)
- Promega RNasin
- Positive control RNA
- Agarose 2%
- Cooler racks for 1.5ml tubes and 0.2ml PCR tubes
- 10µl and 200µl adjustable pipettes and aerosol barrier tips
- 0.2ml PCR tubes, slips, or plates
- Sterile, nuclease-free 1.5ml tubes
- Disposable powder-free gloves
- Microcentrifuge
- Vortex
- PCR thermocycler

Primers sequence

Type/subtype	Gene fragment	Primer	Sequence
Influenza B Victoria lineage	HA	Bvf224 Bvr507	ACATACCCTCGGCAAGAGTTTC TGCTGTTTTGTTGTTGTCGTTTT
Influenza B Yamagata lineage	HA	BYf226 BYr613	ACACCTTCTGCGAAAGCTTCA CATAGAGGTTCTTCATTTGGGTTT

Expected PCR product sizes are 284 bp and 388 bp, respectively.

Procedure

1. Extract viral RNA from clinical specimen with QIAGEN RNeasy Mini Kit or equivalent extraction kit according to manufacturer's instructions.

¹³ WHO Collaborating Centre for Reference and Research on Influenza, Chinese National Influenza Centre. National Institute for Viral Disease Control and Prevention. Chinese Centre for Disease Control and Prevention. 155 Changbai Road, Changping District. 102206 Beijing, China. <http://www.cnic.org.cn>.

2. Prepare master mixture for one step RT-PCR as below:

Reagent	Volume (μl)	Final concentration
PCR buffer	5.0	1
dNTPs	1.0	
Primer: BV HA F	0.5	10μM
Primer: BV HA R	0.5	10μM
Primer: BY HA F	0.5	10μM
Primer: BY HA R	0.5	10μM
Enzyme mix	1.0	
RNase Inhibitor	0.1	
Water (molecular grade)	10.9	
Total	20ul	

3. Dispense 20μl of master mix into each test tube.

4. Add 5μl of purified RNA to the above reaction mix.

5. Set the follow thermal cycling conditions:

Step	Temperature (° C)	Time (minute:second)	No. of cycles
Reverse transcription	60	1:00	1
	42	20:00	
	50	20:00	
Activation	95	15:00	1
Denaturation	94	0:30	35
Annealing	52	0:30	
Extension	72	1:00	
Post-PCR extension	72	10:00	1

6. Agarose gel electrophoresis of RT-PCR products: prepare 2% agarose gel, load PCR products and molecular weight marker, and run according to standard protocols. Visualize presence of marker and PCR product bands under UV light.

Interpretation of results

1. The expected PCR product sizes for B-Victoria lineage is 284 bp and for B-Yamagata lineage is 388 bp.
2. There should be no cross reaction with the other influenza B lineage.

Limitations

1. Analysts should be trained and familiar with testing procedures and interpretation of results prior to performing the assay.
2. A false negative result may occur if inadequate numbers of copies of the virus are present in the specimen due to improper collection, transport, or handling.

Validation of PCR:

The following viruses were used to validate these protocols:

B/Chongqing-Yuzhong/1384/2010 (B-Victoria)

B/Hubei-Wujiagang/158/2009 (B-Yamagata).

This PCR protocol and others are available in English on the CNIC web site at:

<http://www.cnic.org.cn/chn/down/showdown.php?downid=663>

E. Conventional RT-PCR assays for the detection of influenza A(H1N1, H3N2), influenza B and avian influenza A(H5N1) viruses¹⁴

This protocol describes conventional RT-PCR procedures for the detection of:

1. A(H1N1)2009 viruses (H1 and N1 genes)
2. Influenza A(H3N2) viruses (H3 and N2 genes)
3. Influenza B viruses (HA and NA genes)
4. Avian influenza A(H5N1) (H5 and N1 genes)
5. Former seasonal influenza A(H1N1) viruses (H1 and N1 genes)

HA and NA genes are amplified as overlapping halves with the primer sets indicated below. Generated PCR products can be used for diagnosis of influenza and sequencing studies.

Materials required

- QIAamp Viral RNA Mini Kit (QIAGEN®, Cat. No. 52904)
- QIAGEN OneStep RT-PCR kit (QIAGEN®, Cat. No. 210212)
- RNase inhibitor 20U/μl, (Applied Biosystems, Cat. No. N808-0119)
- Ethanol (96–100%)
- Microcentrifuge (adjustable, up to 13 000 rpm)
- Adjustable pipettes (10, 20, 100, 200 μl)
- RNAsin (Promega #N2515)
- SS III RT (Invitrogen #18080-085)
- Pfx Polymerase (Invitrogen #11708-039)
- Water (QIAGEN®, Cat. No. 129114)
- Sterile, RNase-free pipette tips with aerosol barrier
- Vortex
- Microcentrifuge tubes (0.2, 1.5 ml)
- Thermocycler: BIORAD DNA Engine (BIORAD)
- Primer set

Procedure

Follow the manufacturer's instructions and elute RNA in 50μl of the supplied buffer.

Use 5μl RNA in a 50μl one-step RT-PCR reaction for clinical sample extracts or 2μl in a 50μl reaction for grown virus extracts.

For RT-PCR, all reactions are run on a BIORAD DNA Engine using thin-walled tubes with calculated (block) temperature control.

¹⁴ WHO Collaborating Centre for Reference and Research on Influenza. National Institute for Medical Research. The Ridgeway, Mill Hill, London NW7 1AA, England. Email: whocc@nimr.mrc.ac.uk. <http://www.nimr.mrc.ac.uk/wic/>

Primers

Primer sets used for one-step RT-PCR for human influenza surveillance (London WHO CC; May 2011)

Type/subtype	Gene fragment	Primer	Sequence	PCR Product Size (bp)
Influenza A(H1N1)2009	HA-5'(H1)	H1F1 H1R1264	AGCAAAAGCAGGGGAAAATAAAAGC CCTACTGCTGTGAAGTGTGATTC	1264
	HA-3'(H1)	H1F848 HARUc	GCAATGGAAAGAAATGCTGGATCTG ATATCGTCTCGTATTAGTAGAAACAAGGGTGTTTT	945
	NA-5'(N1)	N1F1 N1R1099	AGCAAAAGCAGGAGTTTAAATG CCTATCCAAACACCATTGCCGTAT	1099
	NA-3'(N1)	N1F401 NARUc	GGAATGCAGAACCTTCTTCTTGAC ATATGGTCTCGTATTAGTAGAAACAAGGAGTTTTTT	1073
Influenza A(H3N2)	HA-5'(H3)	H3A1F6 H3A1R1	AAGCAGGGGATAATTCTATTAACC GTCTATCATTCCCTCCCAACCATT	1127
	HA-3'(H3)	H3A1F3 HARUc	TGCATCACTCCAAATGGAAGCATT ATATCGTCTCGTATTAGTAGAAACAAGGGTGTTTT	863
	NA-5'(N2)	NAFUc H3N2R1095	TATTGGTCTCAGGGAGCAAAAGCAGGAGT TCATTTCCATCATCRAAGGCCCA	1095
	NA-3'(N2)	N2F387 NARUc	CATGCGATCCTGACAAGTGTTATC ATATGGTCTCGTATTAGTAGAAACAAGGAGTTTTTT	1082
Influenza A Matrix	Full gene	MF1 MR1027	AGCAAAAGCAGGTAGATATTGAAAGA AGTAGAAACAAGGTAGTTTTTACTC	1027
Influenza B	HA-5'	BHAF1u BHAR1341	TATTCGTCTCAGGGAGCAGAAGCAGAGCATTTTCTAATATC TTCGTTGTGGAGTTCATCCAT	1361
	HA-3'	BHAF458 BHA2R1	AGAAAAGGCACCAGGAGGACCCTA GTAATGGTAACAAGCAAACAAGCA	1391
	NA-5'	BNAF1u BNAR2	TATTCGTCTCAGGGAGCAGAAGCAGAGCATCTTCTCA GATGGACAAATCCTCCCTTGATGC	1130
	NA-3'	BNAF2 BNAR1487	GCACTCCTAATTAGCCCTCATAGA TAAGGACAATTGTTCAAAC	1182
Influenza A(H5N1)	HA-5'(H5)	H5A1F1 H5R1265	AGCAAAAGCAGGGGTATAATC ACGGCCTCAAAGTGTTCATT	1263*
	HA-3'(H5)	H5F417 H5A2R1	TTGAGAAAATWCAGATCATCCC AAGGGTGTTTTTAACTAACAATCT	1351*
	NA-5'(N1)	H5N1F4 H5N1R1112	AGCAAAAGCAGGAGATTAAATGAAT TTCTCCCGATCCAAACACCATTGC	1104
	NA-3'(N1)	H5N1F461 H5N1R1457	GACTGTCAAAGACAGAAGCCCTCA GTAGAAACAAGGAGTTTTTTGAA	997
Former seasonal Influenza A(H1N1)	HA-5'(H1)	THAF2 SPHAR11	GCAGGGGAAAATAAAAACAACC TATTTTGGGCACTCTCCTATTG	990
	HA-3'(H1)	H1HAF552 HARUc	TACCCAAACCTGAGCAAGTCCTAT ATATCGTCTCGTATTAGTAGAAACAAGGGTGTTTT	1239
	NA-5'(N1)	H1N1F6 NASPR10	AGCAGGAGATTAAATGAATCCAA CCTTCCTATCCAAACACCATT	1097
	NA-3'(N1)	N1F741 NARUc	ATAATGACCGATGGCCCGAGTAAT ATATGGTCTCGTATTAGTAGAAACAAGGAGTTTTTT	737

HA and NA genes are amplified as overlapping halves with the primer sets indicated. Generated products can be used for diagnosis of influenza and sequencing studies.

* Size will vary slightly depending on the length of the encoded polybasic (amino acid) cleavage site.

Procedure

One-Step RT-PCR protocol

1. QIAGEN® OneStep RT-PCR kit (Cat. No. 210212) kit is sufficient for 100 x 50µl reactions following manufacturer's instructions.
2. Invitrogen reagent-based one-step protocol used at the National Institute for Medical Research (NIMR, London).

Reagent	Volume (µl) Clinical	Volume (µl) Virus	Final concentration
Water (molecular grade)	30.6	33.6	
10X Buffer*	7.5	7.5	
50mM MgSO ₄ *	1.0	1.0	
100mM dNTPs	0.9	0.9	25mM of each dNTP
10 µmol/l Forward primer [†]	1.5	1.5	0.3µmol/l final concentration
10 µmol/l Reverse primer [†]	1.5	1.5	0.3µmol/l final concentration
RNAsin	0.5	0.5	
SS III RT	1.0	1.0	
Pfx Polymerase*	0.5	0.5	
RNA	5.0	2.0	
Total	50.0	50.0	

* Supplied with the Pfx polymerase.

[†] See above for primer pairings dependent on the type/subtype being amplified.

Thermal cycler BIORAD DNA Engine programme:

Temperature (°C)	Time (minute:second)	No. of cycles
50	30:00	1
94	10:00	1
94	0:05	40
55	0:05	
68	2:00	
68	10:00	1
4	Hold	

Two-step RT-PCR

For weak clinical samples (e.g. those with low Ct values [30 or higher] in qRT-PCR) an 'optimized' two-step protocol is used.

a) RT step

Type	Gene fragment	Primer	Sequence
Influenza A	all genes	uni12W	AGCRAAAGCAGG
Influenza B	all genes	Buni11W	AGCAGAAGCGS

For a 40 µl reaction:

Reagent	Volume (µl)	Final concentration
Water (molecular grade)	12.1	
5X buffer*	8.0	
0.1M DTT *	2.0	
RNAsin	2.0	
100mM dNTPs	0.9	25mM of each dNTP
30µmol/l Uni12W or Bui 11W	3.0	
SS III RT	2.0	
Template RNA	10.0	
Total	40.0	

* Supplied with the SS III RT.

Method

Mix primer and template in a thin-walled tube and incubate at 65° C/5 min. Remove from heat source (DNAEngine) and allow to cool to room temperature. Centrifuge briefly before adding 27µl reaction mix, then mix and briefly centrifuge before thermal cycling using the programme:

Temperature (° C)	Time (minute:second)
25	5:00
50	60:00
70	15:00

b) PCR step

For a 50 µl reaction:

Reagent	Volume (µl)	Final concentration
Water (molecular grade)	35.1	
10X buffer*	7.5	
MgSO ₄ *	1.0	
100mM dNTPs	0.9	25mM of each dNTP
10µmol/l Forward primer [†]	1.5	
10µmol/l Reverse primer [†]	1.5	
Pfx	0.5	
RT product	2.0	
Total	50.0	

* Supplied with the Pfx polymerase.

† See above for primer pairings dependent on the type/subtype being amplified.

Method

Mix primers and template in a thin-walled tube, then add 45µl reaction mix. Mix and briefly centrifuge before thermal cycler programme:

Temperature (° C)	Time (minute:second)	No. of cycles
94	10:00	1
55	5:00	1
68	2:00	1
94	0:05	39
55	0:05	
68	2:00	
94	0:05	1
55	0:05	1
68	10:00	1
4	Hold	

Product analysis

Run 5µl each sample on a 0.8% (w/v) agarose gel made up with 1X TBE buffer and containing GelRed dye (Biotium, Cat. No. 41003-1) according to manufacturer's instructions.

Reactions should yield single bands and do not require gel purification.

Product clean-up

It is necessary to remove RT-PCR component reagents prior to gene sequencing. This is best done using a column DNA-capture/elute process and the system used at NIMR is from GE Healthcare (illustra GFX PCR DNA and gel band purification kit No. 28-9034-70). Manufacturer's instructions are followed, but 2 x 500µl washes are used and, for sequencing purposes, products are usually eluted with either 50µl of water (QIAGEN®, Cat. No. 129114) or the 'pink' elution buffer supplied with the GE Healthcare kit.

Product quantification

Yields of DNA are measured using a GeneQuant pro (Cat. No. 80-2114-98) and the equivalent of 100-200ng of DNA is used per sequencing reaction.

Gene sequencing

Performed using ABI BigDye® Terminator v1.1 Cycle Sequencing kits (Applied Biosystems, Cat. No. 4336774) and capillary based sequencers (MegaBACE 1000 or ABI X3730).

Annex 2: Realtime RT-PCR protocols

Realtime RT-PCR poses different challenges than conventional RT-PCR. In addition to the RT-PCR considerations described in Annex 1, specific considerations for realtime RT-PCR include:

- Ensuring appropriate equipment, software, and fluorescent-based reagents are used and handled correctly.
- Ensuring appropriate training of personnel for interpretation of results (experience in recognizing true positives, interpreting controls/Ct value and aberrant fluorescence is crucial).
- Validation in the laboratory and optimization of reactions are essential to making quantitative determinations.
- There is little likelihood of contamination when reactions are discarded after testing. However, many laboratories do further post-reaction analysis (e.g. restriction fragment length polymorphism using gels, sequencing, etc.) which can re-introduce contamination.

A. Realtime RT-PCR test for the matrix gene of influenza type A viruses¹⁵

Extract viral RNA from clinical specimen as described in Annex 1: Conventional RT-PCR analyses.

Materials required

Reverse transcription

- 10X PCR buffer I with 15mmol/l MgCl₂ (Applied Biosystems Cat No. 4379876)
- Random hexamer 50µmol/l (Applied Biosystems, Cat. No. 8080127)
- MuLV Reverse Transcriptase 50U/µl (Applied Biosystems, Cat. No. 8080018)
- RNase Inhibitor 20U/µl (Applied Biosystems, Cat. No. 8080119)

Realtime PCR

- LightCycler® – FastStart™ DNA Master HybProbe kit (Roche Applied Sciences, Cat. No. 03 003 248 001)

Primers and probes

Working primer and probe mix is prepared by adding equal volumes of the following six components:

Type/subtype	Gene fragment	Primer	Sequence
Influenza type A	Matrix (M)	FLUAM-1F	AAGACCAATCCTGTACCTCTGA (10µmol/l)
		FLUAM-2F	CATTGGGATCTTGCACTTGATATT (10µmol/l)
		FLUAM-1R	CAA AGCGTCTACGCTGCAGTCC (10µmol/l)
		FLUAM-2R	AAACCGTATTTAAGGCGACGATAA (10µmol/l)
		FLUA-1P	5'-(FAM)-TTTGTGTTACAGCTCACCGT-(TAMRA)-3' (5µmol/l)
		FLUA-2P	5'-(FAM)-TGGATTCTTGATCGTCTTTCTTCAAATGCA-(TAMRA)-3' (5µmol/l)

¹⁵ Virology Division, Centre for Health Protection, Hong Kong SAR, China, (National Influenza Centre, WHO H5 Reference Laboratory). <http://www.chp.gov.hk/>

Procedure

1. Prepare master mixture for reverse transcription as below:

Reagent	Volume (μl)
10X PCR buffer I with 15mmol/l MgCl ₂	2.0
Extra 25mmol/l MgCl ₂	2.8
dNTPs (2.5mmol/l)	8.0
Random hexamer 50μmol/l	1.0
RNAase inhibitor 20U/μl	1.0
Reverse transcriptase 50U/μl	1.0
Total volume	15.8

2. Add 4.2μl viral RNA to the above mix.
3. Vortex and centrifuge the tube with the mixture briefly (~3 sec).
4. Stand the tube at room temperature for 10 minutes and then incubate at 42° C for at least 15 minutes.
5. Incubate the tube at 95° C for 5 minutes and then chill in ice.
6. Prepare master mixture for RT-PCR as below:

Reagent	Volume (μl)
Water (molecular grade)	7.6
MgCl ₂ (25mmol/l)	2.4
Primers and probe mix	3.0
'Hot Start' reaction mix *	2.0
Total volume	15.0

* Prepare 'Hot Start' reaction mix according to Roche LightCycler - FastStart DNA Master HybProbes kit's instructions.

7. Add 5μL cDNA to the above mix.
8. Perform realtime RT-PCR according to the following conditions:

Step	Temperature (° C)	Time (minute: second)	No. of cycles
Initial PCR activation	95	10:00	1
Denaturation	95	0:10	50
Annealing	56	0:15	
Extension	72	0:10	
Cooling	40	0:30	1

Data analysis

1. When the run has completed, click Finish.
2. Click on Analysis in the Global Toolbar and select Absolute Qualification of the Analysis type for data analysis.
3. Select channel 530 from channel setting and 640 for channel denominator to read the results.

B. One-step realtime RT-PCR for H1 gene of A(H1N1)2009 virus

Protocol 1¹⁶

This protocol is a realtime RT-PCR to detect A(H1N1)2009 virus (HA gene) in specimens from humans. It is recommended that laboratories having concerns about identifying currently circulating viruses should contact one of the WHO CCs or WHO H5 Reference Laboratories for assistance in identifying the optimal primers to be used.

Materials required

- QIAamp Viral RNA Mini Kit (QIAGEN®, Cat. No. 52904)
- 7500 Real-Time PCR System (Applied Biosystems)
- Invitrogen SuperScript® III Platinum® one-step qRT-PCR System (No. 11732-088)

Primers and probes

Type/subtype	Gene fragment	Primer	Sequence
Influenza A(H1N1)2009	HA	swlH1F swlH1R swlH1P*	GACAAAATAACAAACGAAGCAACTGG GGGAGGCTGGTGTTCATAGCACC GCATTCGCAA"t"GGAAAGAAATGCTGG

* Lower case "t" denotes position of quencher. Probes need to be labeled at the 5'-end with the reporter molecule 6-carboxyfluorescein (FAM) and quenched internally at a modified "t" residue with BHQ1, with a terminal phosphate at the 3'-end to prevent probe extension by DNA polymerase.

Procedure

1. Extract viral RNA from clinical specimen with QIAamp Viral RNA Mini Kit or equivalent extraction kit according to manufacturer's instructions.
2. Prepare master mixture for RT-PCR as below:

Reagent	Volume (µl)	Final Concentration
Water (molecular grade)	5.5	
2X PCR master mix*†	12.5	5X
Forward primer	0.5	40µM
Reverse primer	0.5	40µM
Probe	0.5	10µM
RT/DNA polymerase mix*	0.5	
Total master mix	20.0	
RNA template	5.0	
TOTAL reaction volume	25.0	

* Supplied in the Invitrogen kit.

† ROX reference dye (supplied with the Invitrogen kit) **must** be added to the master mix at the level recommended by the manufacturer.

¹⁶ WHO Collaborating Centre for Reference and Research on Influenza. National Institute for Medical Research, The Ridgeway, Mill Hill, London NW7 1AA, England. Email: whocc@nimr.mrc.ac.uk. <http://www.nimr.mrc.ac.uk/wic/>

3. Assemble a master mix for the required number of samples (remember to make up more than required to account for pipetting losses).
4. Make 20µl aliquots of this and add the required RNA template. Briefly centrifuge the plates/tubes prior to loading the thermal cycler and running the thermal cycler programme.

Thermal cycler amplification programme:

Step	Temperature (° C)	Time (minute:second)	No. of cycles
Reverse transcription and activation of Taq	50	30:00	1
	95	02:00	
PCR	95	00:15	45
	55	00:30*	

* Fluorescence data (FAM) is collected during the 55°C incubation step.

Protocol 2¹⁷

This protocol is a realtime RT-PCR to detect A(H1N1)2009 viruses (HA gene) in specimens from humans. It is recommended that laboratories having concerns about identifying currently circulating viruses should contact one of the members of the WHO Expert Committee on influenza PCR or one of the WHO CCs for assistance in identifying the optimal primers to be used.

Materials required

- QIAamp Viral RNA Mini Kit (QIAGEN®, Cat. No. 52904)
- Ethanol (96–100%)
- Microcentrifuge (adjustable, up to 13 000rpm)
- Adjustable pipettes (10, 20, 100, 200µl)
- Sterile, RNase-free pipette tips with aerosol barrier
- Vortex
- Microcentrifuge tubes (0.2, 1.5ml)
- RealTime ready RNA Virus Master kit (Cat. No. 05619416001 for 100 reactions or 05992877001 for 1000 reactions) (Roche):
 - Enzyme Blend, 50 × conc. (vial 1, red cap)
 - Reaction Buffer, 5 × conc. (vial 2, colorless cap)
 - Water, PCR-grade (vial 3, colorless cap)

¹⁷ Virology Division, Centre for Health Protection, Hong Kong SAR, China, (National Influenza Centre, WHO H5 Reference Laboratory). <http://www.chp.gov.hk/>

Primers and probes

Working primer and probe mix is prepared by adding equal volume of the following three components:

Type/subtype	Gene fragment	Primer	Sequence
Influenza A(H1N1)2009	HA	H1-sw-91F H1-sw-205R H1-sw-119P	GCATAACGGGAAACTATGCAA(10 μ mol/l) GCTTGCTGTGGAGAGTGATTC(10 μ mol/l) 5'-(FAM)-TTACCCAAATGCAATGGGGCTACCCC-(BBQ)-3'(10 μ mol/l)

Procedure

1. Perform RNA extraction of clinical specimens.
2. For each test sample and positive and negative controls, prepare master mixture as below:

Reagent	Volume (μ l)
Master Mix:	
Water (molecular grade)	7.6
5X Reaction buffer	4.0
Primers and probes mix	3.0
Enzyme Blend	0.4
Total volume	15.0

3. Add 5 μ L viral RNA to the above mix.
4. Perform one-step real-time RT-PCR according to the following conditions:

Step	Temperature ($^{\circ}$ C)	Time (minute:second)	No. of cycles
Reverse transcription	50	8:00	1
Initial PCR activation	95	0:30	1
Denaturation	95	0:01	45
Annealing	56	0:20	
Extension	72	0:01	
Cooling	40	0:30	1

Data analysis

1. When the run has completed, click Finish.
2. Click on Analysis in the Global Toolbar and select Absolute Qualification of the Analysis type for data analysis.
3. Select channel 530 from channel setting to read the results.

C. Influenza B lineage specific one-step realtime RT-PCR¹⁸

Materials required

- QIAamp Viral RNA Mini Kit (QIAGEN®, Cat. No. 52904) or equivalent
- Water (molecular grade)
- Qiagen OneStep RT-PCR Kit (QIAGEN®, Cat. No. 210212)
- Primers and probes
- RNase Inhibitor (40U/μl, e.g. TaKaRa Cat. No. 2310A)
- Adjustable pipettes and sterile, RNase-free pipette tips with aerosol barrier
- Real-time thermocycler (method validated for Corbett Rotor Gene RG-3000/RG-6000)
- Positive control virus, Victoria/2/87 lineage, e.g. B/Shangdong/7/1997
- Positive control virus, Yamagata/16/88 lineage, e.g. B/Florida/4/2006

Test validation

The method had been tested against a wide variety of historical and recent influenza B reference viruses, including the lineage prototype strains B/Victoria/2/87 and B/Yamagata/16/88, and in all cases the method has clearly distinguished between the two lineages. The identity of a large number of viruses from patient specimens testing positive with the method has been verified by sequence analysis and antigenic characterization.

Furthermore, the method yielded consistently negative results against various influenza A viruses (former seasonal H1N1; 2009 pandemic H1N1; H3N2; H2N3; H5N1; H7N3; H9N2) as well as various non-influenza respiratory viruses (parainfluenza 1, 2, 3; RSV A and B; adenovirus) and human respiratory specimens that has tested negative for influenza B.

The test has been validated on Corbett Rotor Gene RG-3000 and RG-6000 instruments. Laboratories using different instruments should first critically and carefully examine the cycling conditions as they may *not* perform optimally on other thermocyclers.

Primers and probes

Primers and probes have been designed to detect current influenza B viruses with probes reacting exclusively to B/Yamagata/16/88- or B/Victoria/2/87-lineage viruses, respectively. Since these probes carry different fluorophores, viruses can be discriminated by the colour of fluorescent emission from hydrolyzed probe.

Type/subtype	Gene fragment	Primer	Sequence Degenerate nucleotides are indicated in bold
B	HA	BHA-188F*	AGACCAGAGGGAACTATGCCC
B	HA	BHA-270R**	TCCGGATGTAACAGGTCTGACTT
B(Victoria lineage)	HA	Probe-VIC2	Yakima Yellow-5'-CAGACAAAATGCACGGGGAAHATAACC-3'-BHQ
B(Yamagata lineage)	HA	Probe-YAM2	FAM-5'-CAGRCCAATGTGTGTGGGGAYCACACC-3'-BHQ

* Schweiger et al. 2000 (Journal of Clinical Microbiology 38(4) 1552–1558)

** Watzinger et al. 2004 (Journal of Clinical Microbiology 42(11) 5189–5198)

¹⁸ Department of Virology, Norwegian Institute of Public Health, P.O. Box 4404 Nydalen, N-0403 Oslo, Norway. Email: olav.hungnes@fhi.no

Procedure

Each RT-PCR run should include, in addition to the specimen reactions, at least one negative control reaction and one positive control reaction for each of the targets, i.e. one Victoria-lineage positive control and one Yamagata-lineage positive control.

1. Extract viral RNA from clinical specimen with QIAamp Viral RNA Mini Kit or equivalent extraction kit according to manufacturer's instructions.
2. Prepare master mixture for RT-PCR as below. (Make sure to prepare some excess volume, e.g. corresponding to one extra reaction volume, to make up for dispensing losses.)

Reagent	Volume (μl)	Final concentration
Water (molecular grade)*	13.9	
Qiagen One-Step RT PCR buffer*	5.0	5X
Qiagen dNTP mix*	1.0	10mM each
BHA-188F forward primer, 500nM final	0.5	25 μM
BHA270R reverse primer, 500nM final	0.5	25 μM
probe-VIC2, 200 nM final	0.5	10μM
probe-YAM2, 200 nM final	0.5	10μM
RNAse Inhibitor, e.g. TaKaRa cat 2310A	0.1	ca 40U/μl
Qiagen 1-step RT PCR enzyme mix*	1.0	25X
Total master mix	23.0	
RNA template	2.0	
TOTAL reaction volume	25.0	

* Supplied in the Qiagen® OneStep RT-PCR Kit.

3. Dispense 23μl of master mix into each reaction vessel.
4. Add 2μl of purified specimen RNA to the individual reaction vessels.
5. Set the following thermocycling conditions:

Temperature (° C)	Time (minute:second)	No. of cycles
50° C	30:00	1
95° C	15:00	1
95° C	00:10	45
54° C*	00:40	
72° C	00:20	

* Fluorescence data is collected during the 54° C incubation step for probe-YAM2 (Green) and probe-VIC2 (Yellow). For the RotorGene thermocyclers, 5 sec is added for each fluorescence read, so step time is set to 30 sec + 2x5sec = 40 sec.

Interpretation of results

For a run to be valid, there should be no amplification in the negative control reaction. Each lineage specific control should display a positive reaction for the corresponding lineage, with a Ct value within the expected range established for the particular control material, and no amplification for the other lineage.

Provided that all controls meet stated requirements, a specimen is considered positive for the influenza B lineage for which there is a clear reaction growth curve that crosses the threshold line within 40 cycles.

Similarly, a specimen is considered negative in the influenza B lineage specific test if there is no growth curve crossing the threshold within 40 cycles for any of the influenza B lineages.

D. Realtime RT-PCR assays for the detection of seasonal influenza viruses and H5N1 influenza viruses

Protocol 1¹⁹

May 2015: Modifications from the previous version (highlighted red below) include:

- *design of an new H3h target*
- *detection/lineage of type B viruses (HA)*

This protocol describes procedures for the detection of influenza A viruses (M gene), A(H1N1)pdm09 (H1 and N1 genes), the seasonal A(H1N1) and A(H3N2) viruses (H1h, N1h, H3h, N2h genes) and the Type B viruses (*detection and lineage - HA genes*) by real-time RT-PCR. In case of suspicion of avian infection by the H5 virus, primers and probes are enclosed.

We suggest the following testing strategy:

- RNA extraction
- Amplification in parallel of M, and GAPDH (to assess quality of the specimen and extraction procedure) genes
- In a separate set, amplification of the HA and NA specific genes.

MATERIAL

Kits

QIAamp Viral RNA (QIAGEN mini Kit 50)
SuperScript™ III Platinum® One-Step Quantitative RT-PCR System
Non acetyled BSA 10%

Ref: QIAGEN 52 904
Ref: Invitrogen 11732-020
Ref: Invitrogen P2046

Primers and probes

Name	Sequences	Mers	PCR Pdt	Ref
GAPDH-6Fw	GAAGGTGAAGGTCGGAGT	18		3
GAPDH-231Rv	GAAGATGGTGATGGGATTTC	20	226 bp	3
GAPDH-202Probe(-)	CAAGCTTCCCCTTCTCAGCC [5']Fam [3']BHQ-1	20		3
GRAM/7Fw	CTTCTAACCGAGGTCGAAACGTA	23		2
GRAM/161Rv	GGTGACAGGATTGGTCTTGTCTTTA	25	202 bp	2
GRAM probe/52/+	TCAGGCCCCCTCAAAGCCGAG [5']Fam [3']BHQ-1	21		2
Seasonal viruses				
H1h-678Fw	CACCCCAGAAATAGCCAAAA	20		1
H1h-840Rv	TCCTGATCCAAAGCCTCTAC	20	163 bp	1
H1h-715probe	CAGGAAGGAAGAATCAACTA [5']Fam [3']BHQ-1	20		1
H3h-1070Fw	ATGGTTGGGAGGGAATG	17		1
H3h-1167Rv	TGCTGCTTGAGTGCTT	16	98 bp	1
H3h-1144dProbe	CTGCTGCTTGTCTCTTCCCT [5']Fam [3']BHQ-1	21		1

¹⁹ Unité de Génétique Moléculaire des Virus Respiratoires. Institut Pasteur, 25 rue du Docteur Roux 75724, Paris Cedex 15, France. Email: grippe@pasteur.fr; <http://www.pasteur.fr>

N1h-1134 Fw	TGGATGGACAGATACCGACA	20	142 bp	1
N1h-1275 Rv	CTCAACCCAGAAGCAAGGTC	20		1
N1h-1206 probe	CAGCGGAAGTTTCGTTCAACAT [5']Fam [3']BHQ-1	22		1
N2h-1150b Fw	GTCCAACCCTAAGTCCAA	18	194 bp	1
N2h-1344 Rv	GCCACAAAACACAACAATAC	20		1
N2h-1290 probe	CTTCCCCTTATCAACTCCACA [5']Fam [3']BHQ-1	21		1
<i>Pandemic viruses</i>				
GRswH1-349Fw	GAGCTAAGAGAGCAATTGA	19	253bp	1
GRswH1-601Rv	GTAGATGGATGGTGAATG	18		1
GRswH1-538Probe(-)	TTGCTGAGCTTTGGGTATGA [5']Fam [3']BHQ-1	20		1
	TCCACGCCCTAATGATAA	18		1
GRswN1-975Fw			110 bp	
GRswN1-1084Rv	TTCTCCCTATCCAAACAC	18		1
GRswN1-1045bProbe(-)	ATCCTTTTACTCCATTGCTCC [5']Fam [3']BHQ-1	22		1
<i>Type B viruses + lineage</i>				
HA(B)-444Fw	ACCCTACARAMTTGGAACYTCAGG	24	243 bp	4
HA(B)-524Rv	ACAGCCCAAGCCATTGTTG	19		4
HA(B)-501bProbe(-)	AAATCCGMTTTTYATGGTAG [5']Fam [3']MGBNFQ	20		4
HA(B)-499Probe(-)	ATCCGTTTCCATTGGTAA [5']Vic [3']MGBNFQ	18		4
<i>Avian H5 viruses</i>				
H5-1544Fw	CCGCAGTATTCAGAAGAAGC	20	140 bp	1
H5-1683Rv	AGACCAGCYAYCATGATTGC	20		1
H5d-1638Probe(+)	AGTGCTAGRGAAGCTCGCMAGTGTAG [5']Fam [3']BHQ-1	25		1
H5-1063Fw	TTTATAGAGGGAGGATGG	18	100 bp	1
H5-1162Rv	GAGTGGATTCTTTGTCTG	18		1
H5d-1088Probe(x)	TGGTAGATGGTTGGTATGGG [5']Hex [3']BHQ-1	20		1
N1av-459Fw	GTTTGAGTCTGTTGCTTGGTC	21	190 bp	1
N1av-648Rv	GCCATTTACACATGCACATTGAG	23		1
N1av-493Probe(+)	CATGATGGCAYYAGTTGGTTGACAA [5']Fam [3']BHQ-1	25		1

1/ National Influenza Center (Northern-France), Institut Pasteur, Paris.

2/ National Influenza Center (Southern-France), CHU, Lyon.

3/ Wong et al., 2005, J. Clin. Pathol. 58;276-280

4/ Adapted primers from Biere et al., 2010 J. Clin. Microbiology. 48:1425-1427

NUCLEIC ACID EXTRACTION

RNA is extracted from specimens using the QIAamp Viral RNA kit (QIAGEN Mini Kit 50 ref 52904). RNA extracted from 140 µl of original sample, is eluted in 60 µl of elution buffer.

MIX PREPARATION FOR ALL SEPARATE PRIMER/PROBE COMBINATIONS

All primers and probes described below were validated under the following conditions.

RT-PCR Mix kit:

- Invitrogen Superscript™ III Platinum® One-Step qRT-PCR system (ref: 11732-088)

Real-time PCR equipments:

- LightCycler 480 (96)

Adjustments may be required for the use of other kits or other real-time PCR instruments.

All Primers and probes for the detection were also validated under the following conditions.

RT-PCR Mix kit:

- Invitrogen Superscript™ III Platinum® One-Step qRT-PCR system (ref: 11732-088)

<u>Mix :</u>	Vol (μl)	[final]
H ₂ O PPI :	1.96	
Reaction mix 2X :	10	3 mM Mg
MgSO ₄ (50mM) :	0.24	0.6 mM Mg
Forward Primer (10μM):	1	0.5 μM
Reverse Primer (10μM):	1	0.5 μM
Probe (10μM):	0.4	0.2 μM
SuperscriptIII RT/Platinum Taq Mix :	0.4	
Final Volume:	15 μl	

<u>Mix Type B viruses</u>	Vol (μl)	[final]
H ₂ O PPI :	0.86	
Reaction mix 2X :	10	3 mM Mg
MgSO ₄ (50mM) :	0.24	0.6 mM Mg
Forward Primer (10μM):	1.8	0.9 μM
Reverse Primer (10μM):	1.2	0.6 μM
Probe FAM (10μM):	0.3	0.15 μM
Probe VIC (10μM):	0.2	0.1 μM
SuperscriptIII RT/Platinum Taq Mix :	0.4	
Final Volume:	15 μl	

CONTROLS

Each real-time RT-PCR assay includes in addition of unknown samples :

- Two negative samples bracketing unknown samples during RNA extraction (negative extraction controls)
- Positive controls (in duplicate); when using in vitro synthesized transcripts as controls include five quantification positive controls (in duplicate) including 10⁴, 10³ and 10² copies of *in vitro* synthesized RNA transcripts.
- One negative amplification control.

AMPLIFICATION CYCLES (LIGHTCYCLER SYSTEM)

Reverse transcription	45°C	15 min	x1
Denaturation	95°C	3 min	x1
Amplification	95°C	10 sec	
	55°C	10 sec	
	72°C	20 sec	x50
	40°C	30 sec	x1
Cooling			

SENSITIVITY

For the M real-time RT-PCR

Sensitivity, in terms of 95% hit rate is about 100 copies of RNA genome equivalent per reaction (this amount of target sequences is always detected), the probability to detect lower amounts of virus decreases, but samples containing 10 copies could be detected.

For the H1 and N1 pdm real-time RT-PCR

Sensitivity is comparable to that of the M real-time RT-PCR and comparable to the sensitivity of the CDC kit ($C_p < 36$ for all positive specimens tested so far).

For the H3h and N2h real-time RT-PCR

Sensitivity of the H3h real-time RT-PCR is equivalent to that of the M real-time PCR ($C_p \text{ H3h} \approx C_p \text{ M}$) but the sensitivity of the N2h real-time RT-PCR is lower ($C_p \text{ N2h} \approx C_p \text{ M} + 5 C_p$)

For the H1h and N1h real-time RT-PCR

Sensitivity of the N1h real-time RT-PCR is equivalent to that of the M real-time PCR ($C_p \text{ N1h} \approx C_p \text{ M}$) but the sensitivity of the H1h real-time RT-PCR is lower ($C_p \text{ H1h} \approx C_p \text{ M} + 4 C_p$)

For the H5 and N1 avian real-time RT-PCR

Sensitivity of the H5 real-time RT-PCR is equivalent to that of the M real-time PCR ($C_p \text{ N1h} \approx C_p \text{ M}$) but the sensitivity of the N1a real-time RT-PCR is lower ($C_p \text{ H1h} \approx C_p \text{ M} + 5 C_p$)

SPECIFICITY

For the H1 and N1 pdm real-time RT-PCR

Limited testing so far showed no detection for seasonal influenza viruses (influenza A(H1N1), A(H3N2), B) nor for specimens known to be positive for other respiratory viruses (influenza C, RSV A, B, hBoV, hPIV1,3, hMPV, HRV, enterovirus, adenovirus, CMV, HSV, VZV).

For swine influenza viruses, detection was positive for A/sw/England/117316/86 (classical swine lineage) and negative for A/sw/England/502321/94 (H3N2).

For A(H1N1)pdm09 viruses, detection was positive for A/California/4/2009 as well as for more than 10 specimens positive for the novel A(H1N1)swl virus

NOTE: the H1pdm real-time RT-PCR does not detect the positive control from the CDC kit. *Detection with the N1pdm set of primers is more robust than with the H1pdm set.*

POSITIVE CONTROL FOR M AND GAPDH REAL-TIME RT-PCR

Positive control for M real-time RT-PCR is an *in vitro* transcribed RNA derived from strain A/Paris 650/06(H1N1). The transcript contains the Open Reading Frame of the M gene (from the ATG to nt 982) as negative strand. Each microtube contains 10^{11} copies of target sequences diluted in yeast tRNA, and lyophilised.

Positive control for GAPDH real-time RT-PCR is an *in vitro* transcribed RNA. The transcript contains the Open Reading Frame of the M gene (from nt 6 (ATG = 1) to nt 231) as negative strand. Each microtube contains 10^{11} copies of target sequences diluted in yeast tRNA, and lyophilised.

Reconstitution of transcribed RNA

Add 100 µl of distilled water to obtain a solution at a concentration of 10^9 copies/µl. Store at -80°C.

Dilute in H₂O to prepare a master bank at 2×10^6 copies/µl. Store at -80°C.

From this prepare a working bank of reagent at 2×10^4 copies/µl in order to avoid freeze/thaw cycles. Working tubes may be stored at -20°C for less than one week.

Positive controls are available upon request (grippe@pasteur.fr)

INTERPRETATION OF RESULTS

GAPDH reactions should give a $C_p < 35$; if higher and otherwise negative results are obtained this may result from:

- poor quality of the specimen with insufficient number of cells ; obtain a new specimen for the same patient
- presence of inhibitors; repeat the procedure with dilutions of the extracted RNA (e.g. 1:10, 1:100) and/or repeat RNA extraction.

Positive reactions for M and H1pdm or N1pdm and negative reactions for H1h, N1h, H3h, N2h : confirmed case for A(H1N1)pdm09 virus

Positive reaction for M and negative for H1pdm and N1pdm and for H1h, N1h, H3h, N2h (usually seen for low virus load in specimen); repeat reactions and/or repeat RNA extraction

Positive reaction for M and negative for H1pdm and N1pdm but positive for either N1h, H3h and negative for H1h and N2h (usually seen for low virus load in specimen); infection with seasonal virus; repeat reactions and/or repeat RNA extraction to determine sub-types

Positive reaction for M and for H1pdm and N1pdm and positive reaction for either N1h, H3h may reflect a cross-contamination or a possible co-infection with both the novel A(H1N1)pdm09 virus and a seasonal virus; repeat RNA extraction and repeat reactions with all necessary precautions to avoid cross-contamination.

Negative reactions for M and positive reaction for HA-B confirmed case for type B virus (Yamagata or Victoria lineages)

Protocol 2²⁰

This protocol describes realtime RT-PCR procedures for the detection of:

1. Influenza type A viruses (M gene)
2. A(H1N1)2009 viruses (HA gene)
3. Former seasonal influenza A(H1N1) (HA gene)
4. A(H3N2) viruses (HA gene)
5. A(H5N1) viruses (Clade 1, 2, 3) (HA gene)
6. Influenza type B viruses (NS gene)

Extract viral RNA from clinical specimen as described in Annex 1: Conventional RT-PCR protocols.

Materials required

- QIAGEN® QuantiTect®, Probe RT-PCR kit (Cat. No. 204443)
 - 2 x QuantiTect®, Probe RT-PCR Master Mix
 - QuantiTect®, RT Mix
- RNase-free water
- RNase Inhibitor (Applied Biosystems, Cat. No. N808-0119)
- Primers
- TaqMan® MGB Probe

Test validation

Chromo-4 Real-time PCR Detection system (BioRad)

LightCycler 2.0 (Roche) or LightCycler 480 (Roche)

²⁰ WHO Collaborating Centre for Reference and Research on Influenza and WHO H5 Reference Laboratory at National Institute of Infectious Diseases (NIID). Gakuen 4-7-1, Musashi-Murayama-shi, Tokyo 208-001, Japan.

Primers and probes

Type/subtype	Gene fragment	Primer	Sequence
Influenza type A	Matrix (M)	MP-39-67For	CCMAGGTCGAAACGTAYGTTCTCTCTATC
		MP-183-153Rev	TGACAGRATYGGTCTTGTCTTTAGCCAYTCCA
		MP-96-75ProbeAs	5'-(FAM)-ATYTCGGCTTTGAGGGGGCCTG-(MGB)-3'
Influenza A H1N1 pandemic virus	HA	NIID-swH1 TMPrimer-F1	AGAAAAGAATGTAACAGTAACACACTCTGT
		NIID-swH1 TMPrimer-R1	TGTTTCCACAATGTARGACCAT
		NIID-swH1 Probe2	5'-(FAM)-CAGCCAGCAATRTTRCATTTACC-(MGB)-3'
Former seasonal A(H1N1)	HA	NIID-H1 TMPrimer-F1	CCCAGGGYATTTGCGYGA CTATGAG
		NIID-H1 TMPrimer-R1	CATGATGCTGAYACTCCGGTTACG
		NIID-H1 Probe1	5'-(FAM)-TCTCAAAYGAAGATACTGAACT-(MGB)-3'
A(H3N2) viruses	HA	NIID-H3 TMPrimer-F1	CTATTGGACAATAGTAAAACCGGGRGA
		NIID-H3 TMPrimer-R1	GTCATTGGGRATGCTTCCATTTGG
		NIID-H3 Probe1	5'-(FAM)-AAGTAACCCCKAGGAGCAATTAG-(MGB)-3'
A(H5N1) viruses (Clade 1, 2, 3)	HA	H5HA-205-227v2-For	CGATCTAGAYGGGGTGAARCCTC
		H5HA-326-302v2-Rev	CCTTCTCCACTATGTANGACCATTG
		H5HA-205-227-For(2010) ¹	CGATCTAAATGGAGTGAAGCCTC
		H5HA-326-302-Rev(2010) ¹	CCTTCTCTACTATGTAAGACCATTG
		H5-Probe-239-RVa ²	5'-(FAM)-AGCCAYCCAGCTACRCTACA-(MGB)-3'
		H5-Probe-239-RVb ²	5'-(FAM)-AGCCATCCCGCAACACTACA-(MGB)-3'
Influenza type B	NS	NIID-TypeB TMPrimer-F1	GGAGCAACCAATGCCAC
		NIID-TypeB TMPrimer-R1	GKTAGGCGGTCTTGACCAG
		NIID-TypeB Probe1	5'-(FAM)-ATAAACTTTGAAGCAGGAAT-(MGB)-3'

Procedure

1. Prepare master mixture for realtime RT-PCR as below:

Reagent	Volume (μl)	Final Concentration
Water (molecular grade) ¹	3.65	
2x QuantiTectProbe® RT-PCR Master Mix	12.5	1X
Forward Primer (10μM)	1.5	0.6μM
Reverse Primer (10μM)	1.5	0.6μM
TaqMan MGB Probe (5pmol/μl) ²	0.5	0.1μM
RNase Inhibitor (20U/μl)	0.1	
QuantiTect® RT Mix	0.25	
Total master mix	20.0	
RNA template	5.0	
TOTAL reaction volume	25.0	

¹ For the reaction of H5 detection, the volume of RNase-free water is changed and primers, H5HA-205-227-For(2010) and H5HA-326-302-Rev(2010) are added as follows ;

RNase-free water 2.9μl
H5HA-205-227-For(2010) (10 μM) 0.375μl
H5HA-326-302-Rev(2010) (10 μM) 0.375μl

² For the reaction of H5 detection, a mixture of two probes is used.

H5-Probe-239-RVa 0.375μl
H5-Probe-239-RVb 0.125μl

2. Dispense 20µl of the reaction mixture into each RT-PCR reaction plate.
3. Add 5µl of the sample RNA to the reaction mixture. For control reactions, use 5µl of distilled water for negative control and 5µl of appropriate viral RNAs for positive control.
4. Program the thermal cycler as shown in the table below.
5. Start the realtime RT-PCR program while the RT-PCR reaction plates are still **on ice**.

PCR Temperature-cycling condition: Chromo-4 Real-time PCR Detection system (BioRad)

Temperature (° C)	Time (minute:second)	No. of cycles
50	30:00	1
95	15:00	1
94	0:15	45
56*	1:00	

* Fluorescence data is collected during 56°C incubation step.

PCR Temperature-cycling condition: LightCycler 2.0 (Roche) and LightCycler 480 (Roche)

Temperature (°C)	Time (minute:second)	No. of cycles
50	30:00	1
95	15:00	1
94	0:15 (ramp rate 1.5 °C/sec)	45
56*	1:15	

* Fluorescence data is collected during 56°C incubation step.

Test result interpretation

Each realtime RT-PCR assay includes:

- Positive controls; three or four 10-fold serial diluted each subtype-specific viral RNAs, including minimum detectable quantity
- Negative control; distilled water

When a Cp(Ct) value for a sample is lower than 40, and less than or equal to the Cp(Ct) value of each minimum detectable quantity positive control, the result indicates positive.

Subtyping of HA is determined when both the M gene and the corresponding HA gene are positive at the same time.

Validation of H5 PCR:

The following highly pathogenic H5N1 viruses were found to be positive in the assay:

Clade	Virus
1	A/Vietnam/1194/2004
2.1	A/Indonesia/6/2005
2.1.1	A/chicken/Pekalongan/BPPV4/2003
2.1.2	A/chicken/Pangkal Pinang/BPPV3/2004
2.1.3	A/Indonesia/5/2005
2.1.3.2	A/Indonesia/NIHRD11931/2011
2.2	A/Turkey/12/2006
2.2.1	A/Egypt/321/2007
2.3.2	A/whooper swan/Hokkaido/1/2008
2.3.2.1	A/Hubei/1/2010
2.3.4	A/Anhui/01/2005
2.4	A/chicken/Guangxi/12/2004
2.5	A/chicken/Kyoto/3/2004

Protocol 3²¹

This protocol describes realtime RT-PCR procedures for the detection of different subtypes of influenza A viruses including A(H7N9) and B viruses).

1. Influenza type A viruses (M gene)
2. Influenza type A (H1N1(pdm)09), H3, H5 and H7) viruses (HA gene)
3. Influenza type B viruses (HA gene)

Materials required

- RealTime ready RNA Virus Master kit (Roche)
- Roche LightCycler Version 2.0 (Roche)

²¹ Virology Division, Centre for Health Protection, Hong Kong SAR, China, (National Influenza Centre, WHO H5 Reference Laboratory). <http://www.chp.gov.hk/>

Primers and probes

Type/ subtype	Gene	Name	Sequence	Working concentrati on (μl)	Reference
Influenza type A	Matrix (M)	FLUAM-7-F	CTTCTAACCGAGGTCGAAACGTA	2	[1]
		FLUAM-161-R	GGTGACAGGATTGGTCTTGTCTTTA	2	
		FLUAM-49-P4	YAK-TCAGGCCCCCTCAAAGCCGAG-BBQ	2	
Influenza type B	HA	B84F	AGGGGAGGTCAATGTGACTG	2	[2]
		B185R	GGGCATAGTTTCCCTCTGGT	2	
		B141P-P2	YAK-TTTTGCAAATCTCAAAGGAACA-BHQ1	2	
A(H1N1) pdm09	HA	H1-sw-91F	GCATAACGGGAACTATGCAA	3.3	[3]
		H1-sw-205R	GCTTGCTGTGGAGAGTGATTG	3.3	
		H1-sw-119P-1	FAM-TTACCCAAATGCAATGGGGCTACCCC-BBQ	3.3	
A(H3N2)	HA	H3-239F	ACCCTCAGTGTGATGGCTTCCAAA	2	[2]
		H3-400R	TAAGGGAGGCATAATCCGGCACAT	2	
		H3-342P	FAM-ACGCAGCAAAGCCTACAGCAACTGTT-BHQ1	2	
A(H5N1)	HA	H5-1012F	TGGGTACCACCATAGCAATGAGCA	4	[2]
		H5-1155R	AATTCCCTTCCAACGGCCTCAAAC	4	
		H5-1042P-P2	FAM-TGGGTACGCTGCAGACAAAGAATCCA-BHQ1	4	
A(H7N9)	HA	H7-F	TCACAGCAAATACAGGGAAGAG	3.3	[2]
		H7-R	CCCGAAGCTAAACCAGAGTATC	3.3	
		H7-P	FAM-TGACCCAGTCAAATAAGCAGCGG-BBQ	3.3	

[1] Terrier O, Josset L, Textoris J, et al. 2011. Cellular transcriptional profiling in human lung epithelial cells infected by different subtypes of influenza A viruses reveals an overall down-regulation of the host p53 pathway. *Virology Journal* 8:285.

[2] Virology Division, Public Health Laboratory Services Branch, Centre for Health Protection, Department of Health, Hong Kong SAR, China.

[3] Cheng PK, Wong KK, Mak GC, et al. 2010. Performance of laboratory diagnostics for the detection of influenza A(H1N1)v virus as correlated with the time after symptom onset and viral load. *Journal of Clinical Virology* 47:182-5.

Procedure

Prepare master mixture for realtime RT-PCR as below:

Reagent	Volume (μl)
Master Mix:	
Water (molecular grade)	7.6
5X Reaction buffer	4.0
Primers and probes mix	3.0
Enzyme Blend	0.4
RNA sample	5.0
Total	15

PCR temperature-cycling condition

1. For amplification of (1) influenza A M gene; (2) influenza A H3 HA gene; (3) influenza A H7 HA gene; and (4) influenza B HA gene

Step	Temperature (° C)	Time (minute:second)	No. of cycles
Reverse transcription	50	08:00	1
Initial PCR activation	95	00:30	1
Denaturation	95	00:01	45
Annealing	58	00:20	
Extension	72	00:05	
Cooling	40	00:30	1

2. For amplification of influenza A H1N1pdm09 HA gene:

Step	Temperature (° C)	Time (minute:second)	No. of cycles
Reverse transcription	50	08:00	1
Initial PCR activation	95	00:30	1
Denaturation	95	00:01	45
Annealing	56	00:20	
Extension	72	00:05	
Cooling	40	00:30	1

3. For amplification of influenza A H5 HA gene:

Step	Temperature (° C)	Time (minute:second)	No. of cycles
Reverse transcription	50	08:00	1
Initial PCR activation	95	00:30	1
Denaturation	95	00:10	45
Annealing	56	00:15	
Extension	72	00:10	
Cooling	40	00:30	1

Test result interpretation

Positive results are indicated by a sigmoidal curve when the cycle number was plotted against the fluorescence emission intensity.

Weak positive cases may show positive reaction for influenza A M gene real time RT-PCR and negative reactions for other subtyping real-time RT-PCRs or *vice versa*. Those weak positive cases should be confirmed by other means such as repeat RNA extraction, using a lower elution volume during RNA extraction, obtaining a new specimen from the same patient or repeat testing in multiple wells.

The absence of positive signal may not rule out the influenza infection. Results should be interpreted together with clinical and epidemiological information.

Sensitivity and Specificity

1. Influenza A M gene real time RT-PCR

The influenza A M gene real time RT-PCR was used to test 68 respiratory specimens, where 21 were previously laboratory confirmed to be positive for influenza A either by viral culture or RT-PCR with another primer/probe set²², while the remaining 47 were negative. The sensitivity and specificity for these 68 respiratory specimens were both 100%.

In addition, the specificity of the influenza A M gene real time RT-PCR was assessed by testing a variety of DNA and RNA respiratory viruses: influenza B virus, influenza C virus, adenovirus type types 1–7, respiratory syncytial virus groups A and B, parainfluenza virus types 1–4, rhinovirus, enterovirus, metapneumovirus, and herpes simplex virus types 1 and 2. None of the samples gave a positive signal.

2. Influenza B HA gene real time RT-PCR

The influenza B HA gene realtime RT-PCR was used to test 65 respiratory specimens, where 17 were previously laboratory confirmed to be positive for influenza B either by viral culture or RT-

²² Cheng PK, Wong KK, Mak GC, et al. 2010. Performance of laboratory diagnostics for the detection of influenza A(H1N1)v virus as correlated with the time after symptom onset and viral load. *Journal of Clinical Virology* 47:182-5.

PCR with another primer/probe set²³, while the remaining 48 specimens were negative. The sensitivity and specificity for detecting those 65 respiratory specimens were both 100%.

In addition, the specificity of the influenza B HA gene real time RT-PCR was assessed by testing a variety of DNA and RNA respiratory viruses: pandemic influenza A(H1N1) 2009 virus, influenza A(H3N2) virus, influenza A(H5N1) virus, influenza A(H7N9) virus, influenza A(H9N2) virus, influenza C virus, adenovirus types 1-7, respiratory syncytial virus groups A and B, parainfluenza virus types 1-4, rhinovirus, enterovirus, metapneumovirus and herpes simplex virus types 1 and 2. None of the samples gave a positive signal.

3. Real-time RT-PCRs for HA genes of influenza A H1, H3, H5 and H7

The sensitivities of these four realtime RT-PCRs were shown to be comparable with the influenza A M gene real time RT-PCR by testing 10-fold serially diluted clinical isolates of respective viruses. For the H5 PCR, viruses belonging to clades 1, 2.1, 2.2, 2.3.2.1, and 2.3.4 have been evaluated. None of the realtime RT-PCRs gave a false positive signal for other subtypes.

Acknowledgements

We are grateful for the WHO Collaborating Center for Reference and Research on Influenza, Chinese Centers for Disease Control and Prevention, Beijing, China for sharing influenza A(H7N9) virus isolates for validation of the sensitivity and specificity of the assays.

²³ van Elden LJ, Nijhuis M, Schipper P, et al. 2001. Simultaneous detection of influenza viruses A and B using real-time quantitative PCR. *Journal of Clinical Microbiology* 39:196-200.

E. Realtime RT-PCR assays for human influenza A(H7N9) virus

Protocol 1²⁴

The protocol is a realtime RT-PCR to specifically detect avian influenza A (H7N9) virus using realtime RT-PCR with specific primers and probes targeting the matrix, H7, and N9 genes.

Materials required

- Real-time fluorescence quantitative PCR analysis system
- Bench top centrifuge for 1.5mL Eppendorf tubes
- 10, 200, 1000µL pipettors and plugged tips
- Vortex
- QIAGEN® RNeasy Mini Kit
- AgPath one-step RT-PCR kit
- Primer set
- Other materials: RNase-free 1.5mL eppendorf tubes, RNase-free 0.2mL PCR tubes, powder-free disposables latex glove, goggles, headgear, shoe cover, tips for pipettors, β- thioglycol, 70% alcohol.

Primers and probes

The specific primers and probes for the H7 and N9 genes are summarized in the table below. In addition, the use of a primer and probe targeted M gene and house-keeping gene such a ribonucleoprotein (RNP) is recommended for typing all influenza A virus and internal control in the tests.

Type/ subtype	Gene	Name	Sequence	Note
A (H7N9)	HA	CNIC-H7F	5'-AGAAATGAAATGGCTCCTGTCAA-3'	Primer
		CNIC-H7R	5'-GGTTTTTCTTGTATTTTATATGACTTAG-3'	Primer
		CNIC-H7P	5'FAM-AGATAATGCTGCATTCCCGCAGATG-BHQ1-3'	Probe
	NA	CNIC-N9	5' –TAGCAATGACACACACTAGTCAAT-3'	Primer
		CNIC-N9R	5' –ATTACCTGGATAAGGGTCATTACACT-3'	Primer
		CNIC-N9P	5'FAM- AGACAATCCCCGACCGAATGACCC -BHQ1-3'	Probe
Influenza type A (Flu A)		InfA Forward	5' GACCRATCCTGTACCTCTGA C 3'	Primer
		InfA Reverse	5' AGGGCATTYTGACAAAKCGTCTA3'	Primer
		InfA Probe1	5' FAM-TGC AGT CCT CGC TCA CTG GGC ACG-BHQ1-3'	Probe
RnaseP		RnaseP Forward	5' AGATTTGGACCTGCGAGCG 3'	Primer
		RnaseP Reverse	5' GAGCGGCTGTCTCCACAA GT3'	Primer
		RnaseP Probe1	5'FAM-TTCTGACCTGAA GGCTCTGCGCG-BHQ1-3'	Probe

Note: FluA and RNase primer/probe sets were published from a WHO protocol provided by the US Centers for Disease Control and Prevention, Atlanta, USA.

²⁴ WHO Collaborating Center for Reference and Research on Influenza, Chinese Centers for Disease Control and Prevention, Beijing, China.

Procedure

1. Nucleic acid extraction

The procedure is performed in a BSL-2 biohazard hood in the specimen preparation area according to the manufacturer's instructions. Elution of the RNA using a final volume of 50µL water is recommended.

2. Quality control parameters

Negative control: Sterile water is extracted as a negative control at the same time as the nuclear acid extraction of the other specimens.

Reagent blank control: RNase-free water

Positive control: RNA of the A(H7N9) virus provided.

Internal positive control: RNP is recommended.

3. The reaction system preparation

(1) Thaw the RT-PCR Master Mix, primers, and probes at room temperature in the reagent preparation area of the BSL-2 facility.

(2) Prepare reaction mixture. Different primer pairs and probes should be prepared in the different tubes respectively. For each reaction:

Reagent	Volume (µL)
2× RT-PCR Master Mix	12.5
primer-forward (40µM)	0.5
primer-reverse (40µM)	0.5
Probe (20µM)	0.5
25xRT-PCR enzymes mix	1.0
Template RNA	5.0
RNase Free H ₂ O	5.0
Total volume	25.0

- Aliquot the reaction mixture into 0.2mL PCR tubes or a 96-well PCR plate as 20µL per tube and label clearly.
- Add 5µL of the template RNA for the negative control, test specimens, or positive control into the separate tubes with the reaction mixture in a BSL-2 biohazard hood in the specimen preparation area.
- Load the tubes in the PCR cycler for realtime RT-PCR detection and use the following programme for cycling:

Temperature (° C)	Time (minute: second)	No. of cycles
45	10:00	1
95	10:00	
95	00:15	
60	00:45	
95	00:15	40
60	00:45	

Interpretation of results

The results are determined if the quality controls work.

- (1) The specimen is negative, if the value of Ct is undetectable.
- (2) The specimen is positive, if the Ct value is ≤ 38.0 .
- (3) It is suggested that specimens with a Ct higher than 38 be repeated.

The specimen can be considered positive, if the repeat results are the same as before; i.e. the Ct value is higher than 38. If the repeat Ct is undetectable, the specimen is considered negative.

Criteria for quality control

- (1) The result of the negative control should be negative.
- (2) The Ct value of the positive control should not be more than 28.0.
- (3) Otherwise, the test is invalid.

Troubleshooting

1. False positives may be due to environmental contamination, if there is amplification detected in the negative control and reagent blank control. The unidirectional work flow must be strictly obeyed. The following measures should be taken should there be false positives: ventilate the laboratories, wash and clean the workbench, autoclave centrifuge tubes and tips, and use fresh reagents.
2. RNA degradation should be taken into consideration if the Ct value of the positive control is more than 30. All materials should be RNase-free.

Cautions

1. In order to avoiding nucleic acid cross-contamination, add the negative control to the reaction mixture first, then the specimen, followed by the positive control respectively.
2. Dedicated equipment for each area, including lab coats, pipettors, plugged tips and powder-free disposal latex gloves, are required.
3. Follow the instructions for maintenance of the incubator, PCR cycler, and pipettors. Calibration should be performed every 6 months.

Biosafety

The lysis of the specimen (500 μ L lysis buffer with 200 μ L clinical samples is recommended) should be to be carried out in a BSL-2 facility with BSL-3 level personal protection equipment. Subsequent procedures can be performed in a BSL-2 laboratory which has separate rooms, including a reagent preparation area, specimen preparation area, and amplification/detection area. The DNA-free area is the clean area and the area of amplified DNA is the dirty area. The work flow is from clean to dirty areas.

Protocol Use Limitations

These protocols were optimized using the quantitative one-step probe RT-PCR (AgPath one-step RT-PCR kit) which have been shown to produce comparable results on 96-well format thermocycler systems such as Stratagene QPCR instruments (MX3000® or MX3005®).

Protocol 2²⁵

This quantitative RT-PCR protocol and primer set are designed to detect human A(H7N9) viruses in human specimens. Suitable biosafety precautions should be made for handling suspected human H7 samples. It is recommended that laboratories having concerns about identifying human H7N9 viruses should contact one of the WHO CCs for assistance in identifying the optimal primers to be used.

Materials required

- QIAamp Viral RNA Mini Kit (QIAGEN®, Cat. No. 52906) or equivalent extraction kit
- TaqMan® EZ RT-PCR Core Reagents (Life Technologies, Cat. No. N8080236)
- Ethanol (96–100%)
- MicroAmp Fast Optical 96-well reaction plate (Life Technologies, Part No. 4346906)
- MicroAmp optical adhesive film (Life Technologies, Part No. 4311971)
- Microcentrifuge (adjustable, up to 13 000rpm)
- Adjustable pipettes (10, 20, 100, 200µl)
- Sterile, RNase-free pipette tips with aerosol barrier
- Vortex
- Microcentrifuge tubes (0.5ml and 1.5ml)
- Thermocycler (Life Technologies, ViiA™ 7 Real-Time PCR System or Applied Biosystems 7500)
- Positive control (Available from HKU, e-mail: lmpoon@hkucc.hku.hk)
- Primer set

Primers and probes

Type/ subtype	Gene	Name	Sequence
A(H7N9)	HA	Human H7 for	ATAGATAGCAGGGCAGTTGG
		H7 rev	GATCWATTGCHGAYTGRGTG ¹
		H7 probe	FAM-CCYTCYCCYTGTGCRTTYTG-BHQ1 ¹

1. Journal of Clinical Microbiology. 48:4275

Procedure

1. Extract viral RNA from the clinical specimen by using QIAamp viral RNA mini kit or equivalent extraction kit according to manufacturer's instructions.
2. Prepare master mixture for one-step RT-PCR as below:

²⁵ Centre of Influenza Research, School of Public Health, The University of Hong Kong, Hong Kong SAR, China, (WHO H5 Reference Laboratory)

Reagent per reaction	Volume (μl)
5x Taqman EZ Buffer	5.0
Manganese acetate (25mM)	3.0
dATP (10mM)	0.75
dCTP (10mM)	0.75
dGTP (10mM)	0.75
dUTP (20mm)	0.75
Forward primer (10μM)	0.5
Reverse primer (10μM)	0.5
Probe (10μM)	0.5
rTth DNA Polymerase (2.5U/ml)	1.0
UNG (1U/ml)	0.25
dH ₂ O	10.25
RNA sample	1.0
Total volume	25.0

3. Set the follow RT-PCR conditions:

Temperature (° C)	Time (minute:second)	No. of cycles
50	02:00	1
60	30:00	
95	05:00	
94	00:20	40
60	01:00	

Interpretation of results

Samples containing HA derived from human influenza A(H7N9) virus are expected to be positive in the assay. The absence of a positive signal from a reaction does not necessarily rule out the presence of influenza virus. Results should be interpreted together with the available clinical and epidemiological information.

Remarks

The forward primer is highly specific for the influenza A(H7N9) virus detected in humans and has ≥ 2 mismatches to other avian and equine H7 sequences. The reverse primer and probe, however, can bind to H7 HA of Eurasian lineage. It is therefore important to conduct the test using the indicated RT-PCR condition. In our evaluation, all RNA samples extracted from human A(H7N9) viruses (N=3) were positive in the test. By contrast, all the studied control samples (American H7, N=1; Eurasian H7, N=6; H1-H12, N=1 for each subtype; human respiratory samples from patients with seasonal influenza virus infection, N=5; and human respiratory samples from patients with other respiratory infection, N=5) were negative in the test.

Protocol 3²⁶

This primer set is adapted from a previous published protocol from Tsukamoto and colleagues (*Journal of Clinical Microbiology*. 48:4275) and it is specific for Eurasian H7 HA (except the one from the human H7N9 virus). This assay might help to identify human viral infection caused by other Eurasian H7 viruses. Suspected human H7 cases that are negative for human H7N9 virus might be considered to be tested by this assay. Suitable biosafety precautions should be made for handling suspected H7 samples. It is recommended that laboratories having concerns about identifying human H7N9 viruses should contact one of the WHO CCs for assistance in identifying the optimal primers to be used.

Materials required

- QIAamp Viral RNA Mini Kit (QIAGEN®, Cat. No. 52906) or equivalent extraction kit
- TaqMan® EZ RT-PCR Core Reagents (Life Technologies, Cat No. N8080236)
- Ethanol (96–100%)
- MicroAmp Fast Optical 96-well reaction plate (Life Technologies, Part No. 4346906)
- MicroAmp optical adhesive film (Life Technologies, Part No. 4311971)
- Microcentrifuge (adjustable, up to 13 000rpm)
- Adjustable pipettes (10, 20, 100, 200µl)
- Sterile, RNase-free pipette tips with aerosol barrier
- Vortex
- Microcentrifuge tubes (0.5ml and 1.5ml)
- Thermocycler (ViiA™ 7 Real-Time PCR System)
- Positive control (Available from HKU, e-mail: llmpoon@hkucc.hku.hk)
- Primer set

Primers and probes

Type/ subtype	Gene	Name	Sequence
A (H7N9)	HA	EA H7 for	ATMAATAGCAGRGCARTRGG ¹
		H7 rev	GATCWATTGCHGAYTGRGTG ¹
		H7 probe	FAM-CCYTCYCCYGTGTCRTTYTG-BHQ1 ¹

1. *Journal of Clinical Microbiology*. 48:4275

Procedure

1. Extract viral RNA from the clinical specimens by using the QIAamp viral RNA mini kit or equivalent extraction kit according to manufacturer's instructions.
2. Prepare master mixture for one step RT-PCR as below:

Reagent per reaction	Volume (µl)
5x Taqman EZ Buffer	5
Manganese acetate (25mM)	3

²⁶ Centre of Influenza Research, School of Public Health, The University of Hong Kong, Hong Kong SAR, China, (WHO H5 Reference Laboratory)

dATP (10mM)	0.75
dCTP (10mM)	0.75
dGTP (10mM)	0.75
dUTP (20mm)	0.75
Forward primer (10μM)	0.5
Reverse primer (10μM)	0.5
Probe (10μM)	0.5
rTth DNA Polymerase (2.5U/ml)	1
UNG (1U/ml)	0.25
dH ₂ O	10.25
RNA sample	1
Total volume	25

3. Set the follow RT-PCR conditions:

Temperature (° C)	Time (minute:second)	No. of cycles
50	02:00	1
52	06:00	
95	05:00	
95	00:10	40
52	00:20	
60	00:32	

Interpretation of results

H7, with the exception of human H7N9, viruses of Eurasian lineage are expected to be positive in the test. Samples containing HA derived from human influenza A(H7N9) virus or its closely related lineages are expected to be negative in the assay. The absence of the positive signal from a sample does not necessarily rule out the presence of influenza virus. Results should be interpreted together with the available clinical and epidemiological information.

Remarks

The degenerated forward primer can cross react with H7 HA of Eurasian lineage but it has two mismatches to the human H7N9 virus. The reverse primer and probe can bind to H7 HA of Eurasian lineage, including the one from human H7N9. It is important to conduct the test using the indicated RT-PCR condition. In our evaluation, all RNA samples extracted from human H7N9 viruses (N=3) were negative in the test. By contrast, all Eurasian avian H7 HA (N=6) were positive in the assay. All the control samples (American H7, N=1; H1-H6, N=1 for each subtype; H8-H12, N=1 for each subtype; human respiratory samples from patients with seasonal influenza virus infection, N=5; and human respiratory sample from patients with other respiratory infection, N=5) were negative in the test.

Protocol 4²⁷

Added May 2015

This protocol describes procedures for the detection of A(H7N9) influenza virus (M, H7 & N9 genes).

Suitable biosafety precautions should be used for handling suspected A(H7N9) specimens.

We suggest the following testing strategy:

- RNA extraction
- Amplification in parallel of M, and GAPDH genes (to assess quality of the specimen and extraction procedure)
- In a separate set, amplification of the HA and NA specific genes.

MATERIAL

Kits

QIAamp Viral RNA (QIAGEN mini Kit 50)

Ref: QIAGEN 52 904

SuperScript™ III Platinum® One-Step Quantitative RT-PCR System

Ref: Invitrogen 11732-020

Primers and probes

The sequences from A/Shanghai/1/2013 (H7N9) virus served as template for assay design, i.e. for the design of the sets of primers and probe and for the design of the synthetic gene to be used as control (GISAID accession numbers: H7 EPI439486/ N9 EPI439487).

The table below provides a summary of the specific primers and probes targeting the H7 and N9 genes as well as primers and probes targeting the M gene for typing all influenza A viruses and a house-keeping gene such a GAPDH to be used as internal control in the tests.

Name	Sequences	Mers	PCR Product	Ref
GAPDH-6Fw	GAAGGTGAAGGTCGGAGT	18	226 bp	3
GAPDH-231Rv	GAAGATGGTGATGGGATTTTC	20		3
GAPDH-202Probe(-)	CAAGCTTCCCGTTCTCAGCC [5']Fam [3']BHQ-1	20		3
GRAM/7Fw	CTTCTAACCGAGGTCGAAACGTA	23	154 bp	2
GRAM/161Rv	GGTGACAGGATTGGTCTTGTCTTTA	25		2
GRAM probe/52/+	TCAGGCCCCCTCAAAGCCGAG [5']Fam [3']BHQ-1	21		2
<i>H7N9 influenza virus</i>				
GRavH7-155Fw	AACGAACAAACATCCCCA	18	186 bp	1
GRavH7-340Rv	CTTCATTACGAATTTCCCA	20		1
GRavH7-311Probe(-)	ACATCACTTCCTTCTCGCCTCT [5']Fam [3']BHQ-1	22		1
GRavN9-67Fw	CTCATTGGAATGGCAAACCT	20	153 bp	1
GRavN9-219Rv	TTCCATTGGATGTTGGTGA	20		1
GRavN9-121Probe(+)	TGCAATTGCTCACACTCACA [5']Fam [3']BHQ-1	20		1

References:

1/ National Influenza Center (Northern-France), Institut Pasteur, Paris.

2/ National Influenza Center (Southern-France), CHU, Lyon.

3/ Wong et al., 2005, J. Clin. Pathol. 58;276-280

²⁷ Unité de Génétique Moléculaire des Virus Respiratoires. Institut Pasteur, 25 rue du Docteur Roux 75724, Paris Cedex 15, France. Email: grippe@pasteur.fr; <http://www.pasteur.fr>

NUCLEIC ACID EXTRACTION

RNA is extracted from specimens using the QIAamp Viral RNA kit (QIAGEN Mini Kit 50 ref 52904). RNA extracted from 140 µl of original sample, is eluted into 60 µl of elution buffer.

MIX PREPARATION FOR ALL SEPARATE PRIMER/PROBE COMBINATIONS

All primers and probes described below for the detection of influenza A viruses (M gene), GAPDH and A(H7N9) (H7 and N9 genes) were validated under the following conditions.

RT-PCR Mix kit:

- Invitrogen Superscript™ III Platinum® One-Step qRT-PCR system (ref: 11732-088)

Real-time PCR equipments:

- LightCycler 480
- LightCycler 96

Prepare master mixture for real-time RT-PCR as below.

<u>Mix :</u>	<u>Vol (µl)</u>	<u>[final]</u>
H ₂ O PPI :	1.96	
Reaction mix 2X :	10	3 mM Mg
MgSO ₄ (50mM) :	0.24	0.6 mM Mg
Forward Primer (10 µM):	1	0.5 µM
Reverse Primer (10 µM):	1	0.5 µM
Probe (10 µM):	0.4	0.2 µM
SuperscriptIII RT/Platinum Taq Mix :	0.4	

Final reaction mix volume: 15 µl

15 µl of reaction mix + 5 µl of RNA samples

CONTROLS

Each real-time RT-PCR assay includes in addition of unknown samples :

- Two negative samples bracketing unknown samples during RNA extraction (negative extraction controls)
- Positive controls (in duplicate); when using *in vitro* synthesized transcripts as controls include five quantification positive controls (in duplicate) including 10⁴, 10³ and 10² copies of *in vitro* synthesized RNA transcripts.
- One negative amplification control.

AMPLIFICATION CYCLES (LIGHTCYCLER SYSTEM)

Detection Format:

LightCycler® 480 Instrument: Fam 483-533 / Hex 523-568

LightCycler® 480 II Instrument: Fam 465-510 / Hex 533-580

- 1: Reverse Transcription (RT): transcription of the RNA to cDNA
- 2: Denaturation: sample denaturation and enzyme activation
- 3: Cycling: PCR-amplification of the target DNA
- 4: Cooling: cooling the instrument

Program Step :	RT	Denaturation	Cycling			Cooling
Parameter						
Analysis Mode	None	None	Quantification mode			None
Cycles	1	1	50			1
Target [°C]	45	95	95	55	72	40
Hold [hh:mm:ss]	00:15:00	00:03:00	00:00:10	00:00:10	00:00:20	00:00:30
Ramp Rate [°C/s] 96	4.4	4.4	4.4	2.2	4.4	2.2
Acquisition Mode	None	None	None	None	Single	None

SENSITIVITY

For the M real-time RT-PCR

Sensitivity, in terms of 95% hit rate is about 100 copies of RNA genome equivalent per reaction (this amount of target sequences is always detected), the probability to detect lower amounts of virus decreases, but samples containing 10 copies could be detected.

For the H7 and N9 real-time RT-PCR

Sensitivity was verified using RNA extracted from a high titer virus stock of A/Anhui/ 1/2013(H7N9) virus. Sensitivity of the H7 and N9 real-time RT-PCR is equivalent to that of the M real-time PCR (Cp H7 or Cp N9 ≈ Cp M).

Expected Cp values with control synthetic RNA transcripts:

1E04 copies: Cp ≈ 28

1E03 copies: Cp ≈ 32

1E02 copies: Cp ≈ 36

SPECIFICITY

For the H7 and N9 real-time RT-PCR

Assay specificity was determined using:

- high-titer virus stock solutions of seasonal A(H1N1)pdm09, A(H3N2) and type B (Victoria and Yamagata lineages) influenza viruses, A/Turkey/England/63(H7N3) and A/chicken/HK/G9/97(H9N2) viruses.
- Samples from WHO EQAP-2011 including H5N1 clade 1, 2.1, 2.2 and 2.3.4 samples.
- clinical specimens (source NIC Northern-France) known to be positive for the following viruses:
 - o influenza viruses : H1N1pdm09 (n=9), H3N2 (n=6), type B Victoria lineage (n=6), type B Yamagata lineage (n=12), type C (n=8)
 - o other respiratory viruses: RSV A (n=8), RSV B (n=8), hMPV (n=11), HRV (n=7), hBoV(n=9).

Identities and virus RNA concentrations were re-confirmed by specific real-time RT-PCRs for each virus prior to testing.

To exclude non-specific reactivity of oligonucleotides among each other, all formulations were tested 24 times in parallel with assays containing water and no other nucleic acids except the provided oligonucleotides. In none of these reactions was any positive signal seen.

POSITIVE CONTROL FOR M, GAPDH, H7 AND N9 REAL-TIME RT-PCR

Positive control for M real-time RT-PCR is an *in vitro* transcribed RNA derived from strain A/Paris 650/06(H1N1). The transcript contains the Open Reading Frame of the M gene (from the ATG to nt 982) as negative strand. Each microtube contains 10¹¹ copies of target sequences diluted in yeast tRNA, and lyophilized.

Positive control for GAPDH real-time RT-PCR is an *in vitro* transcribed RNA. The transcript contains the Open Reading Frame of the M gene (from nt 6 (ATG = 1) to nt 231). Each microtube contains 10^{11} copies of target sequences diluted in yeast tRNA, and lyophilized.

Positive control for H7 and N9 real-time RT-PCR is a combined (H7 and N9 genes) *in vitro* transcribed RNA. The transcript contains the Open Reading Frame of the H7 gene (from nt 12 (ATG = 1) to nt 427 and from nt 1528 to nt 1683) and the N9 gene (from nt 12 (ATG = 1) to 272) as negative strand. Each microtube contains 10^{11} copies of target sequences diluted in yeast tRNA, and lyophilized.

Reconstitution of transcribed RNA

Add 100 µl of RNase-free distilled water to obtain a solution at a concentration of 10^9 copies/µl. Store at -80°C.

Dilute in H₂O to prepare a master bank at 2×10^6 copies/µl. Store at -80°C.

From this prepare a working bank of reagent at 2×10^4 copies/µl in order to avoid freeze/thaw cycles. Working tubes may be stored at -20°C for less than one week.

Positive controls are available upon request (grippe@pasteur.fr)

INTERPRETATION OF RESULTS

GAPDH reactions should give a Cp < 35; if higher and otherwise negative results are obtained this may result from:

- poor quality of the specimen with insufficient number of cells ; obtain a new specimen for the same patient
- presence of inhibitors; repeat the procedure with dilutions of the extracted RNA (e.g. 1:10, 1:100) and/or repeat RNA extraction.

Positive reactions for M and H7 or N9 : confirmed case for A(H7N9) virus

Positive reaction for M and negative for H7 and N9 and for either seasonal influenza virus (might be observed for low virus load close to the detection limit); repeat reactions and/or repeat RNA extraction and/or obtain a new specimen for the same patient

Positive reaction for M and negative for H7 and N9 but positive for either seasonal influenza virus; infection with seasonal virus

Positive reaction for M and for H7 and N9 and positive for either seasonal influenza virus: may reflect a cross-contamination or a possible co-infection with both the novel A(H7N9) virus and a seasonal virus; repeat RNA extraction and repeat reactions with all necessary precautions to avoid cross-contamination.

AKNOWLEDGMENTS

We gratefully acknowledge the authors, originating and submitting laboratories of the sequences from GISAID's EpiFlu Database used for assay design (www.gisaid.org).

We are grateful to the originating laboratories for the A/Anhui/1/2013(H7N9) and A/chicken/HK/G9/97(H9N2) isolates used for validation of the sensitivity and specificity of the assay that were shared under the PIP Framework and respectively kindly provided by J. McCauley, WHOCC London and R. Webby, WHOCC Memphis.

F. Realtime RT-PCR assays for human influenza A(H10N8) virus

Protocol 1²⁸

To specifically detect avian influenza virus A(H10N8) virus using real-time RT-PCR with specific primers and probes targeting the matrix, H10 and N8 genes.

Materials required

- Real-time fluorescence quantitative PCR analysis system
- Bench top centrifuge for 1.5mL Eppendorf tubes
- 10, 200, 1000µL pipettors and plugged tips
- Vortex
- QIAGEN® RNeasy Mini Kit
- AgPath one-step RT-PCR kit
- The specific primers and probes for the H10 and N8 genes are summarized in the table below. In addition, the use of a primer and probe targeted M gene and house-keeping gene such as RNP is recommended for typing all influenza A virus and internal control in the tests.
- Other materials: RNase-free 1.5mL eppendorf tubes, RNase-free 0.2mL PCR tubes, powder-free disposables latex glove, goggles, headgear, shoe cover, tips for pipettors, β-thioglycol, 70% alcohol.

Primers and probes

The specific primers and probes for the H7 and N9 genes are summarized in the table below. In addition, the use of a primer and probe targeted M gene and house-keeping gene such as ribonucleoprotein (RNP) is recommended for typing all influenza A virus and internal control in the tests.

Type/ subtype	Gene	Name	Sequence
Influenza type A (H10N8)	HA	CNIC-H10F	5'- GCAGAAGAAGATGGRAAAGGR-3'
		CNIC-H10R	5'-GCTTCCTCTCTGTACTGTGWATG-3'
		CNIC-H10P	5'FAM-TGCATGGAGAGCATMAGAAACAACACCT-BHQ1-3'
	NA	CNIC-N8	5' –AGCTCCATTGTGATGTGTGG-3'
		CNIC-N8R	5' –AGGAAGAATAGCTCCATCGTG-3'
		CNIC-N8P	5'FAM- ACYATGAGATTGCCGACTGGTCA-BHQ1-3'
Influenza type A (FluA)	InfA Forward		5' GACCRATCCTGTACCTCTGA C 3'
	InfA Reverse		5' AGGGCATTYTGGACAAKCGTCTA3'
	InfA Probe1		5' FAM-TGC AGT CCT CGC TCA CTG GGC ACG-BHQ1-3'
RnaseP	RnaseP Forward		5' AGATTGGACCTGCGAGCG 3'
	RnaseP Reverse		5' GAGCGGCTGTCTCCACAA GT3'
	RnaseP Probe1		5'FAM-TTCTGACCTGAA GGCTCTGCGCG-BHQ1-3'

*Note: FluA and RNase primer/probe sets were from published WHO protocol provided by CDC, Atlanta.

²⁸ WHO Collaborating Center for Reference and Research on Influenza, Chinese Centers for Disease Control and Prevention, Beijing, China.

Procedure

1. Nucleic acid extraction

The procedure is performed in a BSL-2 biohazard hood in the specimen preparation area according to the manufacturer. Elution of the RNA using a final volume of 50µL water is recommended.

2. Quality control parameters

Negative control: Sterile water is extracted as a negative control at the same time as the nucleic acid extraction of the other specimens.

Reagent blank control: RNase-free water

Positive control: RNA of the A(H10N8) virus

Internal positive control: RNP is recommended.

3. The reaction system preparation

(1) Thaw the RT-PCR Master Mix, primers, and probes at room temperature in the reagent preparation area of the BSL-2 facility.

(2) Prepare reaction mixture. Different primer pairs and probes should be prepared in the different tubes respectively. For each reaction:

Reagent	Volume (µL)
2× RT-PCR Master Mix	12.5
primer-forward (40µM)	0.5
primer-reverse (40µM)	0.5
Probe (20µM)	0.5
25x RT-PCR enzymes mix	1.0
Template RNA	5.0
RNase Free H ₂ O	5.0
Total volume	25.0

4. Aliquot the reaction mixture into 0.2mL PCR tubes or a 96-well PCR plate as 20µL per tube and label clearly.
5. Add 5µL of the template RNA for the negative control, test specimens, or positive control into the separate tubes with the reaction mixture in a BSL-2 biohazard hood in the specimen preparation area.
6. Load the tubes in the PCR cycler for realtime RT-PCR detection and use the following programme for cycling:

Temperature (° C)	Time (minute: second)	No. of cycles
45	10:00	1
95	10:00	
95	00:15	
60	00:45	
95	00:15	40
60	00:45	

Interpretation of results

The results are determined if the quality controls work.

- (1) The specimen is negative, if the value of Ct is undetectable.
- (2) The specimen is positive, if the Ct value is ≤ 38.0 .
- (3) It is suggested that specimens with a Ct higher than 38 be repeated.

The specimen can be considered positive, if the repeat results are the same as before; i.e. the Ct value is higher than 38. If the repeat Ct is undetectable, the specimen is considered negative.

Criteria for quality control

- (1) The result of the negative control should be negative.
- (2) The Ct value of the positive control should not be more than 28.0.
- (3) Otherwise, the test is invalid.

Troubleshooting

3. False positives may be due to environmental contamination, if there is amplification detected in the negative control and reagent blank control. The unidirectional work flow must be strictly obeyed. The following measures should be taken should there be false positives: ventilate the laboratories, wash and clean the workbench, autoclave centrifuge tubes and tips, and use fresh reagents.
4. RNA degradation should be taken into consideration if the Ct value of the positive control is more than 30. All materials should be RNase-free.

Cautions

4. In order to avoiding nucleic acid cross-contamination, add the negative control to the reaction mixture first, then the specimen, followed by the positive control respectively.
5. Dedicated equipment for each area, including lab coats, pipettors, plugged tips and powder-free disposal latex gloves, are required.
6. Follow the instructions for maintenance of the incubator, PCR cycler, and pipettors. Calibration should be performed every 6 months.

Biosafety

The lysis of the specimen (500 μ L lysis buffer with 200 μ L clinical samples is recommended) should be to be carried out in a BSL-2 facility with BSL-3 level personal protection equipment. Subsequent procedures can be performed in a BSL-2 laboratory which has separate rooms, including a reagent preparation area, specimen preparation area, and amplification/detection area. The DNA-free area is the clean area and the area of amplified DNA is the dirty area. The work flow is from clean to dirty areas.

Protocol Use Limitations

These protocols were optimized using the quantitative one-step probe RT-PCR (AgPath one-step RT-PCR kit) which have been shown to produce comparable results on 96-well format thermocycler systems such as Stratagene QPCR instruments (MX3000® or MX3005®).

Annex 3: Sequencing protocols

Protocol 1²⁹

One-step conventional RT-PCR for amplification and sequencing of the HA and NA genes from avian influenza A(H7N9) virus

Materials Required

- QIAamp Viral RNA Mini Kit (QIAGEN® Cat. No. 52904)
- RNase-free water
- Ethanol (96-100%)
- MyTaq One-Step RT-PCR kit (Bioline BIO-65049)
- Microcentrifuge
- Adjustable pipettes (10, 20, 200, 1000µl)
- Sterile, RNase-free pipet tips with aerosol barrier
- Vortex
- Microcentrifuge tubes (0.2, 1.5ml)
- Thermal cyclers
- Primer sets
- E-gel (Invitrogen)

Primer sequences

Type/subtype	Gene fragment	Primer	Sequence
H7 Fragment I	H7	HAIM13F HAIM13R	TGTAAAACGACGGCCAGTATGAACACTCAAATCCTG CAGGAAACAGCTATGACCACAATTGGCATCAAC
H7 Fragment II	H7	HAIIIM13F HAIIIM13R	TGTAAAACGACGGCCAGTCATAGCTCCAGAC CAGGAAACAGCTATGACCTTATATACAAATAGTGCACC
	N9	NAIM13F NAIM13R	TGTAAAACGACGGCCAGTATGAATCCAAATCAGAAG CAGGAAACAGCTATGACCGTGAACACTACTGG
	N9	NAIIM13F NAIIM13R	TGTAAAACGACGGCCAGTCATACTAAGAACACAG CAGGAAACAGCTATGACCTTAGAGGAAGTACTC

For the H7 gene fragments, the expected product size is 860 bp and 890 bp, respectively.

For the N9 gene fragments, the expected product size is 720 bp and 740 bp, respectively.

Sequencing primers

M13F: TGTAAAACGACGGCCAGT
M13R: CAGGAAACAGCTATGACC

²⁹ RT-PCR protocols may be requested from the WHO CC for Reference and Research on Influenza, Melbourne, Australia.

Procedure

1. Extract viral RNA from clinical specimen using QIAamp viral RNA mini kit or equivalent extraction kit according to manufacturer's instructions.
2. Preparation of master mix for one step RT-PCR:

Reagent	Volume (μl)
2x MyTaq RT-PCR buffer	12.5μl
Forward Primer (20μM)	0.5μl
Reverse Primer (20μM)	0.5μl
RiboSafe RNase Inhibitor	0.5μl
RT/Taq	0.25μl
RNase free water	7.75μl
Total volume	22.0μl/test

Mix gently by pipetting the master mix up and down gently and spin down briefly in a centrifuge.

3. Dispense 22μl of the master mix to each PCR reaction tube.
4. Add 3μl sample RNA to master mix in the PCR tube. Include proper positive control RNA and water for negative control.
5. Start the RT-PCR reaction using the following program:

Temperature (°C)	Time (minute:second)	No. of cycles
45	40:00	1
95	01:00	1
95	00:10	40
60	00:10	
72	01:00	
72	02:00	
7	Hold	Hold

6. Check 8μl PCR products on E-Gel according to manufacturer's instruction, take gel photo with a suitable Gel Doc system.
7. PCR product cleanup with ExoSAP-IT. Add 2μl of PCR product in a PCR tube containing 2μl of ExoSAP-IT and 3μl of H₂O, mix well with gentle pipetting. Place the mixture in thermocycler and incubate at 37° C for 15 minutes, followed by 80° C for 15 minutes.
8. Each PCR products cleaned by ExoSAP-IT is to be sequenced from both ends using M13F and M13R primers. Prepare cycle sequencing master mixes with M13F or M13R primers, add 9μl master mix into corresponding wells of a 96-well PCR plate.

Reagent	Volume for 1 reaction (μl)
BigDye 5x buffer	2

BigDye 3.1 (1:5 dilution)	2
M13F or M13R primer (4µM)	1
Sterile MilliQ water	4
Total	9

9. Add 1µl of purified PCR product (5-20ng) into 9µl of M13F and M13R master mix, pulse spin the plate, place in thermocycle, and start the sequencing reaction using the following conditions:

Temperature (°C)	Time (minute:second)	No. of cycles
96	00:10	25
50	00:05	
60	04:00	
7	Hold	Hold

10. Clean up the sequencing reactions and load onto sequencer according to corresponding standard protocol.

Additional RT-PCR protocols for the detection of influenza viruses and primers for full genome sequencing

Primer sequences for additional RT-PCR protocols and full genome sequencing may also be requested using the following link:

http://www.influenzacentre.org/flucentres_sequencing.htm